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Test – 21

Environment & Ecology

Topics covered:

- ACTS AND POLICIES
- INSTITUTION AND MEASURES
- ENVIRONMENTAL ORGANISATIONS, INTERNATIONAL ENVIRONMENTAL CONVENTIONS
- AGRICULTURE
- Miscellaneous topics
- Current Affairs August 2021



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DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

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TEST BOOKLET SERIES

Test Code:

DPP

TEST BOOKLET
GENERAL STUDIES
TEST - 21

E 21

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERD: **ACTS AND POLICIES, INSTITUTION AND MEASURES, ENVIRONMENTAL ORGANISATIONS
INTERNATIONAL ENVIRONMENTAL CONVENTIONS, AGRICULTURE, Miscellaneous topics,
Current Affairs August 2021**



Q.1) Consider the following statements with reference to Biodiversity Conservation and Rural Livelihood Improvement Project for India:

1. It is funded by the World Bank and implemented by the GEF trust fund.
2. Its aim is to develop and promote new models of conservation at the landscape scale through enhanced capacity and institution building for mainstreaming biodiversity conservation outcomes

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2) Which of the following statements is/are correct with reference to the Nagar Van (Urban Forest) Scheme ?

- a) The Ministry of Environment, Forest and Climate Change (MoEFCC) announced the implementation of the Nagar Van Scheme on the eve of the UN International Day of Forests, 2020.
- b) The Union Ministry of Environment, Forest and Climate Change (EFCC) has selected Itanagar for Urban forest scheme
- c) The Urban forest located in Gujarat will be considered as a role model for this.
- d) Both a and c statements are correct

Q.3) Consider the following statements with respect to the National Neeranchal Watershed Project:

1. The aim of the project is to support the Integrated Watershed Management Program (IWMP) for the improvement and conservation of water resources of India
2. The project is funded jointly by the New Development Bank and Government of India
3. The union Ministry of Agriculture and Farmers Welfare is the implementation agency.

Which of the above statements is/are correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 only
- d) 1, 2 and 3

Q.4) Which of the following statements is not correct with respect to 'Interest Subvention Scheme (ISS) for farmers ?

- a) It aims to provide concessional short term crop loans including loan availed through Kisan Credit Card (KCC) to the farmers.
- b) There are provisions of foregoing interest amount in case of for prompt and timely repayment of loans by the farmers
- c) Animal Husbandry and Fisheries are also included under the ISS.
- d) Reserve Bank of India (RBI) / National Bank for Agriculture and Rural Development (NABARD) is the implementing agency for ISS.

Q.5) Consider the following statements regarding Commission for Air Quality Management (CAQM):

1. It is an executive body.
2. It will complement the functioning of the Environmental Pollution (Prevention and Control) Authority (EPCA).
3. The commission will have an exclusive jurisdiction over the National Capital Region (NCR) in matters of air pollution.

Select the incorrect statement(s) from the code given below.

- a) 1 only.
- b) 1 and 2 only.
- c) 2 and 3 only.
- d) All of the above.

Q.6) With reference to the National Biodiversity Authority (NBA), consider the following statements:

1. It is an autonomous and statutory body.
2. It can take measures necessary to oppose the grant of intellectual property rights in any country outside India on any biological resource obtained from India.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None

Q.7) Consider the following statements regarding National Board for Wildlife (NBWL):

1. Its role is advisory in nature, though it is a statutory organization.
2. Primary function of the Board is to promote the conservation and development of wildlife and forests.



3. No alteration of boundaries in national parks and wildlife sanctuaries can be done without approval of the NBWL.

Which of the statements given above is/are correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.8) The Biodiversity Finance Initiative (BIOFIN) is launched by which of the following organizations ?

- a) International Union for Conservation of Nature.
- b) United Nations Development Program.
- c) World Wide Fund for Nature.
- d) United Nations Educational, Scientific and Cultural Organization.

Q.9) With reference to the Animal Welfare Board of India, consider the following statements:

- 1. It is a statutory body under the Ministry of Environment, Forest and Climate Change.
- 2. The Animal Welfare Board of India was set up under the provisions of National Board for Wildlife in India with the union Minister of Environment, forest and Climate Change as its Chairman.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.10) Which of the following is the function of the Central Zoo Authority ?

- a) To identify endangered species of wild animals for purposes of captive breeding and assigning responsibility in this regard to a zoo.
- b) To advise the Government on any matter connected with animal welfare or the Prevention of infliction of unnecessary pain or suffering on animals.
- c) To ensure maintenance of stud- books of endangered species of wild animals bred in captivity
- d) To coordinate the acquisition, exchange and loaning of animals for breeding purposes.

Q.11) Consider the following statements with reference to 'Biodiversity Samrakshan Internship Programme':

- 1. It is a joint initiative of the National Biodiversity Authority and the United Nations Development Programme.
- 2. The interns selected will have to technically assist the State Biodiversity Boards/UTs Biodiversity Council.

Which of the above mentioned statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.12) "Not All Animal Migrate by Choice" campaign is launched by ?

- a) Wildlife Crime Control Bureau of India and UN Environment
- b) International Union of Conservation of Nature and World Wide Fund for Nature
- c) Wildlife Crime Control Bureau and Interpol
- d) Ministry of Environment Forest & Climate change and Border Security Force.

Q.13) Consider the following statements with reference to Wildlife Crime Control Bureau:

- 1. The Wildlife Crime Control Bureau is a statutory multi-disciplinary body under the MoEFCC.
- 2. X ray images of Wildlife products catalogue were released by WCCB to curb the smuggling of contraband wildlife products.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.14) Which of the following statements is incorrect with respect to the Biological Diversity Act, 2002 ?

- a) The Act was enacted to meet the obligations under the Convention on Biological Diversity (CBD)
- b) The act envisaged a two-tier structure(Centre and State level) to regulate the access to biological resources
- c) The Act exempts endemic resources that are traditionally used as commodities.
- d) None of the above mentioned statements is incorrect.



Q.15) Which among the following are recognised Globally Important Agricultural Heritage Systems sites in India ?

1. Darjeeling Himalayan Tea Estates
2. Kuttanad Below Sea Level Farming System of Kerala.
3. Nagpur Orange Cultivation
4. Koraput Traditional Agriculture of Odisha.
5. Pampore Saffron Heritage of Kashmir.

Select the correct Ans: from the options given below :

- a) 1, 2, 3 and 5 only
- b) 2, 4 and 5 only
- c) 1, 3, 4 and only
- d) 1, 2, 3, 4 and 5

Q.16) Consider the following statements with reference to the High Altitude Tiger(HAT) Project:

1. It is an initiative of the Government of India In collaboration with Global Tiger Forum to assess the population of tigers in high altitude areas.
2. The initiative will track the tiger population concerning the geographical expanse of India and Nepal only.

Which of the above mentioned statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.17) The Union Ministry of Environment Forest and Climate Change recently released the 'Vision 2021-2031 Plan' in reference to ?

- a) Restoring and protecting the Biodiversity of the West Ghats.
- b) Phased introduction of Bio-Fuels for the countries transport sector
- c) Limiting the use of Hydrofluorocarbons as per the provisions of Kyoto Protocol
- d) To upgrade the Zoos in India to global standards and strengthen the Central Zoo Authority.

Q.18) With reference to the Agriculture Infrastructure Fund, consider the following statements:

1. It is a central sector scheme to inject formal credit into farm and farm-processing based activities.
2. Federations of Farmers Producers Organizations (FPOs) and Federations of Self Help Groups (SHGs) are eligible for financing under this scheme

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.19) Which of the following correctly describes the 'Amensalism' type of ecological interaction ?

- a) Secretion of chemicals by a plant that harms the growth or kills neighbouring plants.
- b) Pilot fishes swim along with the sharks and feed on the left overs of the shark's prey.
- c) Nitrogen fixation by fungi.
- d) Pollination and seed dispersal by the insects and birds.

Q.20) Consider the following statements with reference to ' Horticulture Cluster Development Programme (CDP)':

1. It is a central sector programme of the Ministry of Agriculture and Farmers' Welfare aimed at growing and developing identified horticulture clusters to make them globally competitive.
2. Turmeric from West Jaintia Hills (Meghalaya) is one of the identified products for development under this programme at the pilot phase.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.21) In the context of Indian agriculture the terms Kodo, Kutki, Kakun, Kuttu are related to

- a) Types of spices cultivated in the state of Kerala
- b) Varieties of millets cultivated in India.
- c) Methodology of rice cultivation in North East India.
- d) Varieties of rice cultivated below sea level at Kuttanad

Q.22) Which of the following statements is correct with reference to Entomopathogenic fungi ?

- a) Entomopathogenic fungi are a group of fungi living in soil that infect insects by penetrating their cuticle to penetrate their bodies to eventually kill them and feed on them.
- b) The application of entomopathogenic fungi in myco-biocontrol of insects in the agriculture sector is of immense significance because of environmental and food safety concerns.



- c) Entomopathogenic fungi are currently used worldwide for control of a wide range of arthropod pests, particularly pests in protected crops or field crops.
- d) All of the above statements are correct.

Q.23) Recently the Indian Council for Agricultural Research (ICAR) has developed 'Harit Dhara' for the purpose of -

- a) Minimizing soil pollution by using biodegradable manures.
- b) Providing subsidies to the farmers for using solar power in post harvesting of crops.
- c) Minimizing cattle methane emissions by developing anti-methanogenic feed supplement
- d) Roadmap in agricultural revolution for maximizing vegetable production.

Q.24) In the context of Indian agriculture, Ranjit Sub1, Swarna Sub1 and Bahadur Sub1, sometimes seen in the news, are related to ?

- a) Saltwater resistance rice varieties exported by Central Rice Research Institute, to the Southeast Asian countries
- b) Applications developed to monitor rice fields by C-DAC
- c) Flood resistant paddy varieties developed by Indian Agricultural Research Institute and International Rice Research Institute.
- d) Green manure developed by the Indian Agriculture Research Institute.

Q.25) Consider the following statements with respect to Biotech- Kisan:

1. Biotech- Kisan that aims at empowering farmers, especially women farmers is a joint initiative of the Ministry of Agriculture and Farmers Welfare and the Ministry of Rural Development.
2. Biotech-Krishi Innovation Science Application Network (Biotech-KISAN) is being implemented in 15 agro-climatic zones of India in phased manner

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.26) Consider the following statements:

1. Kerala-based Rubber Research Institute of India (RRII) planted the world's first genetically modified (GM) rubber plant near Thiruvananthapuram.
2. Kerala is the largest producer of natural rubber followed by Tamil Nadu in India.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.27) Consider the following statements with respect to 'Operation Greens':

1. The scheme aims for unhindered supply of three crops (TOP): Tomato, Onion and Potato only and the issue related with price of them.
2. The union Ministry of Agriculture and Farmers Welfare implements the scheme.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.28) The Food and Agricultural Organization assisted project named "Technical Assistance for Development of Sustainable Agricultural Value Chains in Mizoram" is related to ?

- a) Cultivation of Birds eye chillies
- b) Increase the production of Ginger in the state.
- c) Assistance to Integrated Crop Livestock Farming Systems
- d) To support local communities in improved jhum management.

Q.29) Which of the following statements correctly defines 'Agricultural yield' ?

- a) Quantity of a crop produced per unit of land.
- b) Total quantity of crop harvested under favourable and controlled conditions.
- c) The weight of grain or other agricultural product, at some agreed standard moisture content, per unit of land area harvested per crop
- d) It is the yield measured in money produced per unit of land



Q.30) Which of the following statements is incorrect with respect to the system of fogponics in agriculture ?

- a) It is a sub-technique of aeroponics where fog is used to create a constant nutrient-rich environment for plant roots.
- b) The plants get the oxygen from the water oxygenated by an air pump in this technique.
- c) The process of fogponics is monitored through Artificial Intelligence induced programming.
- d) None of the above statement

Q.31) Consider the following statements with respect to 'Central Research Institute for Dryland Agriculture (CRIDA)':

- 1. It is a National Research Institute under the Indian Council of Agricultural Research (ICAR) established in 1985 with a mandate to carry out basic and applied research in rainfed farming.
- 2. It prepares contingency plans which prescribe alternate strategies in the event of climate variability in India.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.32) With reference to the "Green – Ag: Transforming Indian Agriculture for global environment benefits and the conservation of critical biodiversity and forest landscapes" project, Which of the following statements is incorrect ?

- a) The project aims to catalyze transformative change for India's agricultural sector to support achievement of national and global environmental benefits and conserve critical biodiversity and forest landscapes.
- b) The Ministry of Agriculture & Farmers Welfare launched the project with collaboration with Food and Agriculture Organization and assisted by the Global Environment Facility (GEF)
- c) Keibul Lamjao National Park of Manipur is designated as one of the conservation-value landscapes under this project.
- d) None of the above statements is incorrect.

Q.33) With reference "Green – Ag: Transforming Indian Agriculture for global environment benefits and the conservation of critical biodiversity and

forest landscapes" project, Which of the following statements is incorrect ?

- a) The project aims to catalyze transformative change for India's agricultural sector to support achievement of national and global environmental benefits and conserve critical biodiversity and forest landscapes.
- b) The Ministry of Agriculture & Farmers Welfare launched the project with collaboration with Food and Agriculture Organization and assisted by the Global Environment Facility (GEF)
- c) Keibul Lamjao National Park of Manipur is designated as one of the conservation-value landscapes under this project.
- d) None of the above statements is incorrect.

Q.34) Consider the following laws related to the environment:

- 1. Indian Forest Act, 1927.
- 2. Wildlife Protection Act, 1972.
- 3. Forest Act (Conservation), 1980.
- 4. Environment (Protection) Act, 1986.
- 5. Public Liability Insurance Act, 1991.
- 6. Biological Diversity Act, 2002.
- 7. Forest Rights Act, 2006

Which of the above acts have some bearing on the authority of NGT?

- a) 1, 3, 4, 5 and 6 only.
- b) 3, 4, 5, 6 and 7 only.
- c) 3, 4, 5 and 6 only
- d) 2, 3, 4 and 6 only

Q.35) With reference to the Environment Protection Act, 1986, consider the following statements:

- 1. The genesis of the act is in Article 48A and Article 51A (g) of the Indian Constitution.
- 2. The act strengthened the *locus standi* rule, empowering common citizens to approach the Court.
- 3. It grants immunity to the officers of the Government for any act done under the provisions of this Act.
- 4. The Act empowers the Civil Courts to entertain any suit or proceeding in respect of an action or order issued by the Central Government.

Select the correct Ans: using the codes given below:

- a) 1, 2 and 3
- b) 1, 3 and 4
- c) 1 and 3



d) 2, 3 and 4

Q.36) With reference to India, consider the following environmental acts:

1. Water (Prevention and Control of Pollution) Act, 1974.
2. Air (Prevention and Control of Pollution) Act, 1981.
3. Forest Conservation Act, 1980.
4. Environment (Protection) Act, 1986.
5. Forest Rights Act, 2006.
6. Water (Prevention and Control of Pollution) Cess Act of 1977.

Which among the above acts empowers the Central Pollution Control Board (CPCB)?

- a) 1, 2, 3, 4 and 6
- b) 2, 3, 4 and 5
- c) 1, 2, 4 and 6
- d) 1, 2, 3 and 6

Q.37) Consider the following pollutants:

1. PM 10
2. Arsenic
3. Lead
4. Nickel
5. Benzene
6. CO
7. Ammonia

Which of the above pollutants are common to both the National Air Quality Index (AQI) and National Ambient Air Quality Standards (NAAQS)?

- a) 1, 3, 4, 5 and 6
- b) 1, 2, 3, 4, 5 and 6
- c) 2, 3, 4, 5 and 6
- d) 1, 3, 6 and 7

Q.38) Consider the following statements with respect to Coastal Regulation Zone Notification, 2018:

1. Rural Areas that are densely populated with a population density below 2161/km² are designated as CRZ-III-A, while those with density more than 2161/km² are designated as CRZ-III-B.

2. No tourism facilities are permissible in the "No Development Zone" (NDZ) of the CRZ-III areas.
3. Floor Space Index (FSI), or the Floor Area Ratio (FAR) had been frozen as per 1991 Development Control Regulation (DCR) levels.

Which of the statement(s) given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) None

Q.39) Consider the following statements with reference to the The Environment (Protection) Act (EPA):

1. The EPA Act can be invoked in case of damage to a property due to hazardous substances.
2. The administration and authority under this act is divided among the Central and State Governments.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.40) Which of the following statements is correct with reference to the features of the E-waste (Management) Amendment Rules, 2018 ?

- a) Producer Responsibility Organizations (PROs) shall apply to the Central Pollution Control Board (CPCB) for registration to undertake activities prescribed in the Rules.
- b) Under the Reduction of Hazardous Substances (RoHS) provisions, cost for sampling and testing shall be borne by the government for conducting the RoHS test for the products that comply with the RoHS provisions
- c) Separate e-waste collection targets have been drafted for new producers, i.e. those producers whose number of years of sales operation is less than the average lives of their products
- d) All of the above statements are correct.

Q.41) "The Critical Wildlife Habitats (CWHs)" are defined as areas that are required to be kept as inviolate for the purposes of wildlife conservation. They are so defined under which of the following acts ?



- a) Scheduled Tribes And Other Traditional Forest Dwellers Act, 2006
- b) Wildlife Protection Act, 1972
- c) Environment Protection Act, 1986
- d) Prevention of Cruelty to Animals Act, 1960

Q.42) With reference to the Forest (Conservation) Act, 1980, consider the following statements:

1. The act clearly defines the term "deemed forests" and makes provisions for their governance.
2. The act stipulates that prior approval of the Central Government is essential for diversion of forest areas for non-forestry purposes by State Governments.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.43) "The Management Effectiveness Evaluation" framework, sometimes seen in the news, is associated with -

- a) Management of Inter-state Rivers in India
- b) Management of protected areas such as national parks and wildlife sanctuaries
- c) Survey of Endemic species in biodiversity hotspots in India
- d) Management of Ground-Water resources in India

Q.44) According to Wetlands (Conservation and Management) Rules, 2017, which of the following are included under the definition of Wetland in India ?

1. Natural Peatland
2. Inundated paddy fields
3. Artificial tanks for aquaculture.
4. Lagoons and creeks
5. Marshes and Fens
6. Artificial Tanks constructed for irrigation purposes

Select the correct Ans: using the code given below:

- a) 1,2,4 and 5 only
- b) 1,4 and 5 only
- c) 2,3,4 and 6 only
- d) 1,3,5 and 6 only

Q.45) In India, The 'Blue Vanda', 'ladies slipper orchids' and the 'Beddome's Cycad', sometimes seen in news are -

- a) Prohibited from being cultivated under all circumstances.
- b) Accorded the same protection as reef building corals.
- c) Cultivated only after an official license is obtained to the same.
- d) Considered as invasive and harmful to the ecosystem.

Q.46) According to the Biological Diversity Act, 2002, which of the following constitute the definition of biological resources under the act ?

1. Genetic material of Animals
2. Value added products derived from plants
3. Human genetic material
4. Genetic Material of Microorganisms

Select the correct Ans: using the code given below:

- a) 1 and 4 only
- b) 1,2 and 4 only
- c) 2 and 3 only
- d) 1,2,3 and 4

Q.47) Which state government became the first Indian state to adopt the Fly Ash Utilization Policy to generate wealth from waste and environment protection ?

- a) Karnataka
- b) Maharashtra
- c) Kerala
- d) Gujrat

Q.48) Recently NASA 's "Lucy Mission "was in the news sometimes, considering the following Statement about it:

1. It's NASA's first spacecraft to study Jupiter's Trojan asteroids.
2. Its mission is to investigate the group of rocky bodies circling the Sun in two swarms.

Which of the above statements is /are correct ?

- a) only 1
- b) only 2
- c) both 1 and 2
- d) D) neither 1 nor 2

Q.49) With reference to the Rashtriya Mahila Kosh, consider the following Statements :

1. It is an autonomous body under the Ministry of finance for socio-economic empowerment of women.



2. Rashtriya Mahila Kosh provides loan products to Non – Governmental Organization (NGO)/ Intermediary Micro-financing Organization (IMO) / Voluntary Organisation (VO) which on-lend to Women Groups like SHG.

Which of the statements given above is / are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 and 2

Q.50) consider the following statement regarding Plastic Waste Management Amendment Rules, 2021:

- 1. According to the amendment it prohibits single use plastic items by 2022.
- 2. Extended Producer Responsibility being brought out has been given legal force through this amendment rules.
- 3. This will not allow reuse of plastic carry bags due to any condition.

Which of the statements given above is/are correct ?

- a) 1 and 2 Only
- b) 2 and 3 Only
- c) 3 Only
- d) All of the above.

Q.51) With reference to the recently launched 'Ubharte Sitaare fund', Which of the following statements is/are not correct ?

- 1. It is an alternative investment fund to help future Champions among export-oriented firms and start-ups.
- 2. It is set up by RBI and Small Industries Development Bank of India (SIDBI).

Select the correct Ans: using the codes given below:

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q. 52) Which of the following is/are Globally climate Tipping point/points ?

- 1. Western Ghats of India
- 2. Amazon Rainforest
- 3. Wilkes Basin
- 4. Coral Reef
- 5. Indian Ocean Circulation
- 6. Atlantic Circulation

Select the correct Ans: using the codes given below:

- a) Only 1, 2, 3 and 4
- b) Only 2, 3, 4, 5 and 6
- c) Only 2, 3, 4 and 6
- d) All of the above.

Q. 53) Which of the following statement best describe e-RUPI:

- a) It is a cashless and contactless instrument for digital payment.
- b) It is a digital currency which is proposed by the RBI.
- c) It is a fintech firm with only a digital presence.
- d) None of the above.

Q.54) Which of the following statements is/are correct regarding MALABAR-21 exercise ?

- 1. It is a military exercise between India, Japan, USA and Australia.
- 2. In 2000 the first ever Malabar exercise was held.

Select the Correct Ans: using the codes given below:

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.55) Consider the following statements with reference to the Deep Ocean Mission :



1. The Ministry of Shipping will be the nodal Ministry for implementing this mission.
2. It has a component for studying and preparing detailed engineering design for offshore Ocean Thermal Energy Conversion (OTEC) powered desalination plants.
3. The Mission will enable India to develop capabilities to exploit resources in the Central Indian Ocean Basin (CIOB).

Which of the statements given above is/ are correct?

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.56) Consider the following statements with regard to the Industrial deep Decarbonisation:

1. It is an United Nation Environment Programme initiative.
2. It helps to tackle carbon intensive construction materials such as steel and cement.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.57) Consider the following statements with regard to the equalization levy:

1. Equalization levy applies only to digital advertisement services and not to e-commerce sellers of goods and services.
2. It is introduced as a part of Income tax Act.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.58) Consider the following statements regarding to the National Human Rights Commission (NHRC) :

1. It is established as a statutory body for protection and promotion of human rights.
2. It has both civil and Criminal power for investigation of human rights issues.
3. The chairman of the Human Rights Commission is appointed by the Prime Minister of India.

Which of the statements given above is/ are correct?

- a) 1 only
- b) 2 and 3 only

- c) 3 only
- d) 1, 2 and 3

Q.59) World Employment and Social Outlook 2021 report was recently released by -

- a) Organisation for Economic Co-operation and Development
- b) International Labour Organisation
- c) World Economic Forum
- d) World Bank

Q.60) Consider the following statements with reference to the One Nation One Standard Mission

1. The purpose of this mission is to converge multiple standards to ensure quality products in the country.
2. This mission is implemented by the Bureau of Indian Standard.
3. Research Design & Standards Organization becomes the First Institution to be declared as Standard Developing Organization under this mission.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.61) Consider the following statements with regard to the 'Jhanda Flag Satyagraha:

1. It was held in Bihar and Bombay in 1923.
2. Sardar Patel, Jamnalal Bajaj, C.Rajagopalachari, Dr. Rajendra Prasad and Vinoba Bhave were prominent leaders to organise the satyagraha.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.62) Consider the following statements with regard to the National Maritime Heritage Complex :

1. The National Maritime Heritage Complex is to be developed at the site of Lotha which will preserve India's rich and diverse maritime heritage.
2. The Ministry of Culture and Ministry of Ports, Shipping and Waterways signed a MoU for Cooperation in Development of National Maritime Heritage Complex.

Which of the statements given above is/are correct?

- a) 1 only



- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.63) Consider the following statements with reference to the Inland Vessels Bill, 2021:

1. Inland Vessels Bill, 2021 will replace the Inland Vessels Act, 1917 to streamline and regulate vessels travel in the inland water fields of India.
2. The Bill provides for a central database for recording the details of vessel, vessel registration, crew on an electronic portal.
3. The bill will help to promote the development of India's inland water transport in a unified manner.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.64) Consider the following statements with reference to the National Monetisation pipeline:

1. National Monetisation pipeline (NMP) is launched in an effort to list out the government's infrastructure assets to be sold over the next ten years.
2. The top three sectors identified for asset monetisation include railways, airports and coal mining

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.65) Consider the following statements with reference to the Foreign Exchange Reserves:

1. Foreign Exchange Reserves, also known as Forex Reserves, are assets held on reserve by a central bank in foreign currencies.
2. Reserve Bank of India is responsible for management of Forex reserves under Reserve Bank of India Act, 1934 and Foreign Exchange Management Act, 1999.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.66) Which of the following statements is incorrect with respect to the Plastic Waste Management Amendment Rules, 2021?

- a) Plastic Waste Management Amendment Rules, 2021 amend the 2016 rules.
- b) The manufacture, import, stocking, distribution, sale and use of the identified single-use plastic will be prohibited with effect from the 1st July, 2022.
- c) Plastic bags with lower thickness are more easily handled as waste and have higher recyclability.
- d) Penalties are imposed under the Environmental Protection Act, 1986.

Q.67) Consider the following statements with reference to the 'Forum for Decarbonizing Transport:

1. The Forum for Decarbonizing Transport will help in achieving specific results in reducing emissions from the transport sector.
2. Ministry of Road Transport and Highways launched the 'Forum for Decarbonizing Transport'.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.68) The term 'surangam' used in the context of -

- a) Water harvesting.
- b) Climate change.
- c) Paintings.
- d) Coinage.

Q.69) With reference to National Heart Failure Biobank (NHFB), consider the following Statements :

1. It is the first National Heart Failure Biobank (NHFB) in the country that would collect blood, biopsies, and clinical data as a guide to future therapies.
2. The biosamples include the blood, serum, tissue samples obtained during open-heart surgery and peripheral blood mononuclear cells (PBMCs) and genomic DNA collected from heart failure patients.
3. The biobank activity is supervised by a Committee headed under the chairmanship of the health minister.

Which of the statements given above is / are correct ?

- a) 1 and 2 only
- b) 2 only



- c) 1, 2 and 3
- d) 1 and 3 only

Q.70) Pensilungpa Glacier Often in news , is situated at -

- a) Spiti Valley in Himachal Pradesh
- b) Zaskar Valley in Ladakh
- c) Dzuko Valley of Nagaland
- d) valley of flowers in uttrakhand

Q.71) Which of the following is correct about stake networking ?

- a) It is made of bamboo, and is installed on the glacier surface using the steam drill for mass balance measurement.
- b) It is the practice of transporting and exchanging data between nodes over a shared medium in an information system.
- c) It is a type of financial network based on stock price correlation used for observing, analyzing and predicting the stock market dynamics.
- d) It is a direct selling business model in which salespeople work independently promoting products or services.

Q.72) Consider the following Statements in the context , Regional Cooperation Agreement on Combating Piracy and Armed Robbery against Ships in Asia (ReCAAP).

- 1. It is the first regional government-to-government agreement to promote and enhance cooperation against piracy and armed robbery against ships in Asia.
- 2. Presently, about 21 countries are members of ReCAAP and all are Asian countries only.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.73) Recently in the news, PUSA- FSF developed by Indian Agricultural Research Institute (IARI), is related to ?

- a) High Yielding Variety of fodder variety for cattles that can grow in minimum water requirement.
- b) Mechanism for in-situ management of paddy stubble for Delhi -NCR region.
- c) The on-farm green energy refrigeration system.
- d) Funds for agricultural focused start- ups.

Q.74) Consider the following statements with reference to Rashtriya Krishi Vikas Yojana – Remunerative Approaches for Agriculture and Allied Sectors Rejuvenation (RKVY-RAFTAAR) scheme:

- 1. It is aimed at strengthening infrastructure in Agriculture and Allied sectors to promote Agripreneurship and Agribusiness by facilitating financial aid and nurturing a system of business incubation.
- 2. Funds granted under the (RKVY-RAFTAAR) scheme are directly transferred to the beneficiaries bank account.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2.

Q.75) Which of the following statements is incorrect with respect to the recently in news Atmanirbhar Bharat ARISE- ANIC (ARISE-Atal New India Challenges) initiative?

- a) It is an initiative launched under the Atmanirbhar Bharat to fast track the growth of the Indian MSME sector.
- b) The Atmanirbhar Bharat ARISE- ANIC initiative is launched by the Atal Innovation Mission (AIM), NITI Aayog and driven by the Indian Space Research Organization as one of the institutions.
- c) Startups can also apply under the Atmanirbhar Bharat ARISE- ANIC initiative
- d) None of the above statements are incorrect.

Q.76) Consider the following statements with reference to the Biotechnology Industry Research Assistance Council (BIRAC):

- 1. The BIRAC is a not-for-profit Public Sector Enterprise, set up by the Department of Biotechnology (DBT), Government of India.
- 2. The National Biopharma Mission (NBM) is implemented by the Biotechnology Industry Research Assistance Council (BIRAC).

Which of the above statements is/are correctly mentioned ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.77) Consider the following statements with reference to the 'Certified Jute Seeds Programme':



1. The Union Ministry of Small and Medium Enterprises has launched a Certified Jute Seed Distribution Plan under the Improved Cultivation and Advanced Retting Exercise for Jute (Jute ICARE) Program.
2. The Improved Cultivation and Advanced Retting Exercise for Jute (Jute – ICARE) was launched in West Bengal and Assam with an aim to double the jute farmers income.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.78) Which of the following statements is not correct with respect to the Protection of Plant Varieties and Farmers' Rights (PPV&FR) Act ?

- a) The PPV&FR Act was enacted to grant intellectual property rights to plant breeders, researchers and farmers who have developed any new or extant plant varieties.
- b) The PPV & FR Act is based on the provisions of the International Treaty of Plant Genetic Resources for Food and Agriculture (ITPGRFA)
- c) The rights granted under this Act are heritable and assignable and only registration of a plant variety confers the right.
- d) None of the above statements are incorrect.

Q.79) Consider the following statements with reference to the 'National Bee Board' (NBB):

1. NBB works under the aegis of the Union Ministry of Medium and Small Enterprises in collaboration with Ministry of Tribal Affairs, aims at preserving the traditional way of Beekeeping and honey production in India
2. The Union Minister of Medium and Small Enterprises is the ex- officio chairman of the NBB .

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.80) Which of the following statements is not correctly mentioned with respect to the 'Maitri Incubation Programme' ?

- a) The Pusa Krishi Incubator, of ICAR- Indian Agricultural Research Institute launched the Maitri Incubation Programme.

- b) It is a six Months Mentoring Program promoting agri start-ups with innovative ideas between India & other BRICS countries.
- c) Only agritech startups were selected on the basis of Startup India guidelines under this programme and will be mentored by experts along with access to global markets.
- d) None of the above statements are incorrect.

Q.81) Consider the following statements with reference to the 'Madhukranti Portal':

1. Madhukranti portal is an initiative of the National Bee Board under the National Beekeeping & Honey Mission (NBHM) with an objective to maintain the quality of honey and also to prevent its adulteration.
2. The technical partner for development of this digital platform is Indian Bank.

Which of the above statements is/are correctly mentioned?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.82) Consider the following statements with reference to the 'Comprehensive Environment Pollution Index (CEPI)':

1. The CEPI is released by the Indian Pollution Control Association in collaboration with Niti Aayog.
2. The index monitors the various air pollutants only and suggests their levels of hazardness.

Which of the above mentioned statements is/are incorrectly mentioned ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.83) Which of the following products are included under the 'Eco Mark' scheme ?

1. Soaps
2. Detergents
3. Paints
4. Textiles
5. Fire extinguisher
6. Leather and coir

Select the correct Ans: from the codes given below:

- a) 1, 2, 5, and 6 only
- b) 2, 3, 4 and 5 only
- c) 4 and 6 only



- d) 1, 2, 3, 4, 5 and 6

Q.84) Consider the following statements with respect to 'Urban Services Environmental Rating System' (USERS) ?

1. USERS is aimed at improving basic services like water supply, sewerage and solid waste.
2. It is funded by the United Nations Development Programme and executed by the Ministry of Environment and Forest and Climate Change.

Which of the above statements is/are correctly mentioned ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.85) Consider the following statements with respect to the 'PUSA KRISHI' mobile app:

1. Pusa Krishi Mobile App disseminates information about the latest technologies developed by the Indian Agricultural Research Institute.
2. Pusa Krishi Mobile App is available in 9 languages viz. English, Hindi, Tamil, Gujarati, Oriya, Punjabi, Marathi, Bangla and Telugu

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.86) Which of the following statements is incorrect with respect to the State of Climate Services report 2021?

- a) The report is released by World Meteorological Organization (WMO)
- b) It focuses on Terrestrial Water Storage.
- c) The biggest losses of Terrestrial Water Storage have occurred in Antarctica and Greenland.
- d) In India the Terrestrial Water Storage has been lost at a rate of at least 5 cm per year.

Q.87) Recently, Noble prize in economics awarded for empirical contributions to:

- a) Labour economics
- b) Auction theory
- c) Developmental economics
- d) Environmental economics

Q.88) Who among the following is associated with the philosophy of Total Revolution?

- a) Bal Gangadhar Tilak
- b) Jawaharlal Nehru
- c) Mohandas Karamchand Gandhi
- d) Jay Prakash Narayan

Q.89) Which of the following statements is incorrect with respect to 'International Energy Agency' ?

- a) The International Energy Agency is a Paris-based autonomous nongovernmental organisation.
- b) It was established under the framework of the Organisation for Economic Co-operation and Development (OECD) in 1974 in the wake of the 1973 oil crisis.
- c) It is best known for the publication of its annual World Energy Outlook.
- d) India became an Associate member of the International Energy Agency.

Q.90) Which of the following statements is incorrect with respect to 'plastic waste recycling rules' ?

- a) Plastic packaging, as per the rules divided into three categories.
- b) Producers of plastic would be obliged to declare to the government how much plastic they produce annually.
- c) If entities cannot fulfil their obligations, they would on a "case basis by case" be permitted to buy certificates making up for their shortfall.
- d) Non-compliance would invite a traditional fine instead of environmental compensation.

Q.91) Which of the following statements is not correct with reference to 'Fluorinated Gases'?

- a) Fluorinated gases are man-made gases and not found naturally.
- b) They have a very long atmospheric lifespan and are high in Global Warming Potential.
- c) Fluorinated gases can be removed from the atmosphere only by destruction with the help of Sunlight.
- d) Nitrogen trifluoride (NF₃) is an example of fluorinated gas.

Q.92) Which of the following statements is incorrectly mentioned about the mechanism 'Climate Forcings'?



- a) Climate forcing is the physical process of affecting the climate system on the Earth through a number of forcing factor
- b) The factors responsible for the climate forcings exists outside of the existing climate_system
- c) The climate system includes the hydrosphere, land surface, and the cryosphere.
- d) None of the above statements are incorrect.

Q.93) Consider the following statements with reference to Global Warming Potential (GWP):

- 1. Global Warming potential describes the impact of the Greenhouse gases only on global warming.
- 2. Carbon dioxide with a GWP of '1' serves as a baseline for other GWP values.

Which of the above mentioned statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.94) Consider the following statements:

- 1. Greenhouse Gas Intensity is a measure of the number of emissions relative to GDP
- 2. The United States of America tops the chart followed by China for Greenhouse Gas Intensity.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.95) Consider the following statements with respect to 'Cumulative Emissions'

- 1. Cumulative emissions is the total quantity of GHG emissions emitted from 1751 until a given year, measured in metric tons.
- 2. The European Union tops the chart followed by the United States of America.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.96) Which of the following kinds of wastes are regulated by the Basel Convention?

- 1. Biomedical Wastes
- 2. Used oils of motors and hydraulics
- 3. Lead Acid Batteries wastes
- 4. Persistent Organic Pollutants
- 5. Chemical wastes generated from industries

Select the correct Ans: from the options given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 5 only
- c) 3 and 5 only
- d) 1, 2, 3, 4 and 5

Q.97) Which of the following statements correctly defines the mechanism of Eutrophication?

- a) The process when an organism absorbs a substance at a rate faster than that at which the substance is lost or eliminated by catabolism and excretion.
- b) It is the accumulation of a chemical by an organism from water and food exposure that results in a concentration that is greater than would have resulted from water exposure only and thus greater than expected from equilibrium.
- c) The increased availability of one or more limiting growth factors and progressive enrichment with minerals and nutrients.
- d) All of the above

Q.98) Which of the following statements is not correctly mentioned with respect to the Acid rain?

- a) The sulphur dioxide and nitrogen oxides react with water, oxygen and other chemicals to form sulfuric and nitric acids which are the precursor of acid rain.
- b) Acid rain can occur at a place away from the polluted area with the source of emissions



- c) Acid rain usually has a pH between 4.2 and 4.4.
- d) The phenomenon of acid rain is entirely due to anthropogenic activities.

Q.99) Consider the following statements regarding the "Basel Convention ":

1. The agreement particularly focuses on preventing transfer of hazardous waste from developed to less developed countries.
2. It is a legally binding agreement for waste management.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.100) "Torrefaction Technology" often seen news is related to -

- a) Technology to reduce stubble burning.
- b) Technology to convert municipal waste into wealth
- c) Technology used in formation of GM crop
- d) Technology used to create energy from tides



Q.1) Ans: B

Exp:

Statement 1 is incorrect & 2nd is correct

- **Biodiversity Conservation and Rural Livelihood Improvement Project for India** aims to develop and promote new models of conservation at the landscape scale through enhanced capacity and institution building for mainstreaming biodiversity conservation outcomes. **The Funding Source is the GEF Trust Fund whereas the Implementing Agency is the World Bank** and the Executing Agency is Government of India, Ministry of Environment and Forests (Central Level), as per India's scenario
- There are four components to the project.
 - **The first component** of the project is demonstration of landscape conservation approaches in two pilot sites: this component will develop tools, techniques, knowledge and skills towards improved conservation and rural livelihoods outcomes in the little Rann of Kutch in Gujarat and Askot in Uttarakhand;
 - **The second component** to the project is to strengthen knowledge management and national capacity for landscape conservation: this component will support improved knowledge and capacity building on learning and experience from the two demonstration landscapes and other local conservation models;
 - **The third component** of the project is to scaling up and replication of successful models of conservation in additional landscape sites: this component would support the further testing and replication of landscape conservation approaches to two additional high biodiversity landscapes from the third year onwards with project financing; and
 - **The fourth component** of the project is national coordination for landscape conservation: this component will support coordination for landscape conservation at the Ministry of Environment and Forests (MOEF).

Source: <https://www.thegef.org/project/biodiversity-conservation-and-rural-livelihoods-improvement>

Q.2) Ans: B

Exp:

Nagar Van (Urban Forest) Scheme

- The Ministry of Environment, Forest and Climate Change (MoEFCC) announced the implementation of Nagar Van Scheme on the eve of the World Environment Day (5th June).
- It aims to develop 200 Urban Forests across the country in the next five years with a renewed focus on people's participation and collaboration between the Forest Department, Municipal bodies, NGOs, Corporates and local citizens.
- The Warje Urban forest located in Pune in Maharashtra will be considered as a role model for this.
- The Scheme will be implemented through people's participation and collaboration between the forest Department, Municipal bodies, NGOs, Corporates and local citizens and the finances for the scheme will be borne by the CAMPA (Compensatory Afforestation Fund (CAF) Act, 2016) funds.
- The Union Ministry of Environment, Forest and Climate Change (EFCC) has selected Arunachal Pradesh capital, Itanagar for implementation of the 'Nagar Van' or Urban Forest scheme.

Source: <https://www.thehindu.com/sci-tech/energy-and-environment/itanagar-selected-for-urban-forest-scheme/article32376369.ece>

Q.3) Ans: C

Exp:

- **Statement 1 is correct:** The Neeranchal National Watershed Project (NNWP) is a World Bank assisted project undertaken by the Government of India to support the Integrated Watershed Management Program (IWMP) for the improvement and conservation of water resources. The project started in 2016 and will continue until 2022
- **Statement 2 is incorrect:** The project will be financed through Investment Project Financing (IPF) at a 50/50 cost-sharing ratio. The total cost of the project is Rs. 2142.30 crore of which the Government's share is Rs. 1071.15 crore (50 percent) and rest is the loan component from the World Bank
- **Statement 3 is incorrect:** The Ministry of Rural Development will implement the project and support the Pradhan Mantri Krishi Sinchayi Yojana in water management. Gujarat, Madhya Pradesh,



Chhattisgarh, Rajasthan, Telangana, Andhra Pradesh, Jharkhand, and Maharashtra are the

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- A **watershed** is the area of land where all of the water that drains off of it goes into the same place—a river, stream or lake. The smallest watersheds are the drainage areas for small streams and lakes.
- The **World Bank** is an international financial institution that provides loans and grants to the governments of low- and middle-income countries for the purpose of pursuing capital projects.

- chosen states for project implementation.

Source:

<https://pib.gov.in/newsite/printrelease.aspx?relid=128501>

Q.4) Ans: B

Exp:

The Interest Subvention Scheme for farmers

- The Interest Subvention Scheme (ISS) was launched in the year 2006-07 with a view to provide concessional short term crop loans including loans availed through Kisan Credit Card (KCC) to the farmers.
- ISS is available to farmers availing short term agri loans upto Rs.3.00 lakh at an interest rate of 7% per annum for one year. The short term agri loans include Crop Husbandry, Animal Husbandry, Dairying and Fisheries. Additional 3% subvention is also given to farmers for prompt and timely repayment of loans; the same gets reduced to 4% per annum.
- The benefits of ISS can also be availed by farmers in allied activities like Animal Husbandry and Fisheries and those farmers who already possess KCC within the overall limit of Rs.3.00 lakhs.
- New KCC can be issued to Animal Husbandry and Fisheries farmers with provision of benefit of IS & PRI for loan amount upto Rs.2.00 lakh per annum.
- The Interest Subvention Scheme is being implemented by NABARD and RBI.

Q.5) Ans: B

Exp:

- Central government has promulgated an ordinance to set up a **Commission for Air Quality Management (CAQM) in the National Capital Region (NCR) and Adjoining Areas** to tackle the air pollution.

- **Statement 1 is incorrect:** CAQM is a **statutory** mechanism and it is a **permanent** body with over 20 members.
- **Statement 2 is incorrect:** CAQM will **replace** the 22-year-old **EPCA**. The EPCA was criticized for not exercising its statutory powers and for merely functioning as an advisory body to the Supreme Court.
- **Statement 3 is correct:** The body will have **exclusive jurisdiction** over the NCR, including areas in Haryana, Punjab, Uttar Pradesh and Rajasthan, in matters of air pollution, and will be working along with CPCB and ISRO, apart from the respective state governments.

Q.6) Ans: C

Exp:

- **Statement 1 is Correct:** National Biodiversity Authority, established in 2003 to implement the Biological Diversity Act, 2002, is an autonomous and statutory body to perform regulatory and advisory functions for Government of India on issues of conservation of biodiversity, sustainable use of its components and realization of fair and equitable sharing of benefits arising out of the use of biological resources.
- **Statement 2 is Correct:** The National Biodiversity Authority may, on behalf of the Central Government, take any measures necessary to oppose the grant of intellectual property rights in any country outside India on any biological resource obtained from India or knowledge associated with such biological resource which is derived from India.

Q.7) Ans: D

Exp:

- **Statement 1 is correct:** NBWL is a “Statutory Organization” constituted under the Wildlife Protection Act, 1972. Its role is “advisory” in nature and advises the Central Government on framing policies and measures for conservation of wildlife in the country.
- **Statement 2 is correct:** Primary function of the Board is to promote the conservation and development of wildlife and forests. It has power to review all wildlife-related matters and approve projects in and around national parks and sanctuaries.



- **Statement 2 is correct:** The WPA, 1972 mandates that without the approval/recommendation of the NBWL, construction of tourist lodges, alteration of the boundaries of PAs, destruction or diversion of wildlife habitat and de-notification of Tiger Reserves, cannot be done.

Q.8) Ans: B

Exp:

- **UNDP in October 2012 launched the Biodiversity Finance Initiative** – BIOFIN, as a new global partnership seeking to address the biodiversity finance challenge in a comprehensive manner – building a sound business case for increased investment in the management of ecosystems and biodiversity, with a particular focus on the needs and transformational opportunities at the national level.
- BIOFIN aims to develop a methodology for quantifying the biodiversity finance gap at national level, for improving cost-effectiveness through mainstreaming of biodiversity into national development and sectoral planning, and for developing comprehensive national resource mobilization strategies.

Q.9) Ans: D

Exp:

- **Statement 1 is incorrect:** The Animal Welfare Board of India, headquartered at Ballabgarh in Haryana state, is a statutory advisory body advising the Government of India's Ministry of Fisheries, Animal Husbandry and Dairying.
- **Statement 2 is incorrect** : The Animal Welfare Board of India, the first of its kind to be established by any Government in the world, was set up in 1962, in accordance with Section 4 of the Prevention of Cruelty to Animals Acts 1960. **Shrimati Rukmini Devi Arundale** pioneered the setting up of the Board, with its Headquarters at Chennai. She guided the activities of the Board for nearly twenty years till her demise in 1986. The Chairman is nominated by the ministry and currently is an Indian Forest Service Officer.

Source: <http://www.awbi.in/about.html>

Q.10) Ans: B

Exp:

The Central Zoo Authority

- The Central Zoo Authority has been constituted under section **38A of the Wild Life (Protection) Act 1972**. The Authority consists of a Chairman, ten members and a Member Secretary.
- The main objective of the authority is to complement the national effort in conservation of wildlife. Standards and norms for housing, upkeep, health care and overall management of animals in zoos have been laid down under the Recognition of Zoo Rules, 1992.
- Every zoo in the country is required to obtain recognition from the Authority for its operation. The Authority evaluates the zoos with reference to the parameters prescribed under the Rules and grants recognition accordingly. Zoos which have no potential to come up to the prescribed standards and norms may be refused recognition and asked to close down
- To advise the Government on any matter connected with animal welfare or the Prevention of infliction of unnecessary pain or suffering on animals is the function of The Animal Welfare Board of India.
- The major functions of the Central Zoo Authority are as follows:
 - To specify the minimum standards for housing, upkeep and veterinary care of animals kept in the zoo.
 - To evaluate and assess the functioning of the zoos with respect to the standard and the norms prescribed
 - To recognize and de recognize zoos
 - To coordinate training of zoo personnel in India and abroad
 - To provide technical and other assistance to zoos for their proper management and development on scientific lines.

Q.11) Ans: C

Exp:

- **Statement 1 is correct** : On the occasion of the International Day of Biodiversity the Union Environment Minister launched the National Biodiversity Authority (NBA) and United Nations Development Programme (UNDP) 'Biodiversity Samrakshan Internship Programme' which proposes to engage 20 students with postgraduate degrees for a period of one year through an open, transparent, online competitive process.
- **Statement 2 is correct** : The programme wishes to engage dynamic and creative students, who are

willing to learn about natural resource management and biodiversity conservation and to support the projects of NBA in various State and Union Territories and to technically assist the State Biodiversity Boards/UTs Biodiversity Council in discharge of their mandates. an open, transparent, online competitive process.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1626054>

Q.12) Ans: A

Exp:

"Not All Animal Migrate by Choice"

- The UN Environment India and Wildlife Crime Control Bureau (WCCB) of India launched an awareness campaign 'Not all animals migrate by choice' to be displayed at major airports across the country.
- In the first phase of the campaign, Tiger, Pangolin, Star Tortoise and Tokay Gecko have been chosen as they are highly endangered due to illegal trading in International markets.
 - Tiger is traded for its skin, bones and body parts; Pangolin, the most illegally traded wild mammal on the planet is trafficked for its meat and its scales are used in traditional medicines.
 - Star Tortoise for meat and pet trade and Tokay Gecko in traditional medicine, mostly into SouthEast Asia and particularly Chinese Markets. Phase two will see more threatened species and explore other routes of trafficking.
- In collaboration with the Airports Authority of India and GMR Group, the campaign will travel across 22 airports across India over the next year. Both WCCB and UN Environment initiated a comprehensive approach with a focus on awareness building of various stakeholders towards the issue of prevention of illegal trade and smuggling of wildlife and wildlife products through exit points.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1572277>

Q.13) Ans: C

Exp:

- **Statement 1 is correct** : Wildlife Crime Control Bureau is a statutory multi-disciplinary body established by the Government of India under the Ministry of Environment and Forests, to combat organized wildlife crime in the country. The Bureau has its headquarter in New Delhi and five regional offices at Delhi, Kolkata, Mumbai, Chennai and Jabalpur; three sub-regional offices at Guwahati, Amritsar and Cochin; and five border units at Ramanathapuram, Gorakhpur, Motihari, Nathula and Moreh. It derives its powers and functions from Section 38 (Z) of the Wild Life (Protection) Act, 1972.
- **Statement 2 is correct**: 'X ray images of Wildlife Products' is a catalogue of Wildlife Contraband Commonly carried in personal baggage by smugglers. The aim of the WCCB in releasing this catalogue is to prevent the organized crime of smuggling and educate the security personnels across the country regarding the various contraband items.

Source: <http://wccb.gov.in/Content/aboutUs.aspx> and <http://wccb.gov.in/WriteReadData/userfiles/file/X%20Ray%20Manual.pdf>

Q.14) Ans: B

Exp:

The Biological Diversity Act, 2002

- The Biological Diversity Act, 2002 is an Act enacted by the Parliament of India for the preservation of biological diversity in India, and provides a mechanism for equitable sharing of benefits arising out of the use of traditional biological resources and knowledge. The Act was enacted to meet the obligations under the Convention on Biological Diversity (CBD), because India is a party of the convention
- The act envisaged a three-tier structure to regulate the access to biological resources:
 - The National Biodiversity Authority (NBA) at the central level
 - The State Biodiversity Boards (SBBs) at the state level
 - The Biodiversity Management Committees (BMCs) at the local level
- Notwithstanding anything contained in this Act, the Central Government may, in consultation with the National Biodiversity Authority, by notification in the Official Gazette, declare that the provisions of this Act shall not apply to any items, including



biological resources normally traded as commodities.

Source:

http://www.jkforest.gov.in/act/THE_BIOLOGICAL_DIVERSITY_ACT.pdf

Q.15) Ans: B

Exp:

"Globally Important Agricultural Heritage Systems"

- The Food and Agriculture Organization (FAO) of the United Nations launched a Global Partnership Initiative on conservation and adaptive management of "Globally Important Agricultural Heritage Systems".
- "Globally Important Agricultural Heritage Systems" (GIAHS) are outstanding landscapes of aesthetic beauty that combine agricultural biodiversity, resilient ecosystems and a valuable cultural heritage. Located in specific sites around the world, they sustainably provide multiple goods and services, food and livelihood security for millions of small-scale farmers
- The overall goal of the GIAHS Programme is to identify and safeguard Globally Important Agricultural Heritage Systems and their associated landscapes, agricultural biodiversity, knowledge systems, and culture.

The three GIAHS in India are:

- Koraput Traditional Agriculture, Odisha
- Kuttanad Below Sea Level Farming System,
- KeralaPampore Saffron Heritage, Jammu & Kashmir

Source: Shankar IAS, Page 315

Q.16) Ans: D

Exp:

- **Statement 1 is incorrect:** The GTF, in partnership with the Governments of Bhutan, India and Nepal, along with technical implementing agencies, including WWF, Wildlife Institute of India, National Trust for Nature Conservation (Nepal) and Regional Centre for Tiger and Cats Conservation (Bhutan) is undertaking a situation analysis study for assessing tiger habitat status in high altitude ecosystems. The study has been supported by the Integrated Tiger Habitat Conservation Program (ITHCP) of the IUCN and KfW.
- **Statement 2 is incorrect:** High altitude tiger (HAT) project will study and analyze the status of Tiger

Habitats in High Altitude Ecosystems in Bhutan, India and Nepal

Source: <https://globaltigerforum.org/projects/high-altitude-tiger-project/>

Q.17) Ans: D

Exp:

- **Vision Plan (2021-2031) for Indian zoos** to upgrade them to global standards and strengthening of Central Zoo Authority- The vision document is committed to making CZA and Indian zoos a greater force for conservation by providing unparalleled animal care, cutting edge research, and immersive visitor experiences that strike meaningful chords with people of all ages.
- The exhaustive 10 years vision plan has been arrived at after a very stringent data mining and stakeholder consultative process and is expected to give a direction towards ex-situ conservation approaches in India.

Source: <https://pib.gov.in/PressReleaseframePage.aspx?PRID=1763045>

Q.18) Ans: C

Exp:

- **Statement 1 is correct :**The Agriculture Infrastructure Fund is a medium - long term debt financing facility for investment in viable projects for post-harvest management infrastructure and community farming assets through interest subvention and credit guarantee.
 - The duration of the scheme is from FY2020 to FY2029 (10 years). Under the scheme, Rs. 1 Lakh Crore will be provided by banks and financial institutions as loans with interest subvention of 3% per annum and credit guarantee coverage under CGTMSE for loans up to Rs. 2 Crores.
 - Eligible beneficiaries include farmers, FPOs, PACS, Marketing Cooperative Societies, SHGs, Joint Liability Groups (JLG), Multipurpose Cooperative Societies, Agri-entrepreneurs, Start-ups, and Central/State agency or Local Body sponsored Public-Private Partnership Projects.
- **Statement 2 is correct :**The Union Cabinet chaired by Hon'ble Prime Minister made the following modifications in Central Sector Scheme of Financing Facility under 'Agriculture Infrastructure Fund'

- Eligibility has now been extended to State Agencies/APMCs, National & State Federations of Cooperatives, Federations of Farmers Producers Organizations (FPOs) and Federations of Self Help Groups (SHGs)
- At present Interest subvention for a loan upto Rs. 2 crore in one location is eligible under the scheme. In case, one eligible entity puts up projects in different locations then all such projects will be now be eligible for interest subvention for loan upto Rs. 2 crore. However, for a private sector entity there will be a limit of a maximum of 25 such projects. This limitation of 25 projects will not be applicable to state agencies, national and state federations of cooperatives, federations of FPOs and federation of SHGs. Location will mean physical boundary of a village or town having a distinct LGD (Local Government Directory) code. Each of such projects should be in a location having a separate LGD code.
- For APMCs, interest subvention for a loan upto Rs. 2 crore will be provided for each project of different infrastructure types e.g. cold storage, sorting, grading and assaying units, silos, et within the same market yard.
- The power has been delegated to Hon'ble Minister of Agriculture & Farmers Welfare to make necessary changes with regard to addition or deletion of beneficiary in such a manner so that basic spirit of the scheme is not altere.
- The period of financial facility has been extended from 4 to 6 years upto 2025-26 and overall period of the scheme has been extended from 10 to 13 upto 2032-33.
- **Ammensalism:** Interaction where one impedes or restricts the success of the other without being positively or negatively affected by the presence of another. The bread mold *Penicillium* is a common example of this; *penicillium* secrete penicillin, a chemical that kills bacteria. A second example is the black walnut tree (*Juglans nigra*), which secretes juglone, a chemical that harms or kills some species of neighbouring plants from its roots.
- **Commensalism:** It occurs when one organism takes benefits by interacting with another organism by which the host organism is not affected. A good example is the pilot fish living with a shark. The shark is not affected in the process as pilot fish only eats the leftover food of the shark which does not deplete the shark's resources.
- **Neutralism:** It is the relationship between two species which do interact but do not affect each other. It is to describe interactions where the fitness of one species has absolutely no effect whatsoever on that of another. True neutralism is extremely unlikely and impossible to prove.
- **Competition:** It is an interaction between individuals or populations that is mutually detrimental.
- **Predation:** Interaction between organisms in which one organism captures biomass from another.
- **Mutualism:** It is an interaction between two or more species, where species derive a mutual benefit. Examples include cleaner fish, pollination and seed dispersal, gut flora and nitrogen fixation by fungi.

Q.20) Ans: C

Exp:

Horticulture Cluster Development Programme

- **Statement 1 is correct:** Horticulture Cluster Development Programme (CDP) is a central sector programme implemented by the National Horticulture Board (NHB) of the Ministry of Agriculture and Farmers' Welfare. It aims at growing and developing identified horticulture clusters to make them globally competitive. In a pilot phase, the programme will be implemented in 12 horticulture clusters out of the total 53 clusters selected for the programme.
- **Statement 2 is correct:** The programme is designed to leverage geographical specialisation and promote integrated and market-led development of horticulture clusters. The Ministry of Agriculture and Farmers' Welfare (MoA&FW) has identified 53

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1733828>:

Q.19) Ans: A

Exp:

Ecological interactions are the relationships between two species in an ecosystem. These relationships can be categorized into many different classes of interactions based either on the effects or on the mechanism of the interaction. The various types of ecological interactions are as follows:



horticulture clusters, of which 12 have been selected for the pilot launch of the programme.

The clusters of the pilot phase include Shopian (J&K) and Kinnaur (H.P.) for Apple, Lucknow (U.P.), Kutch (Gujarat) and Mahbubnagar (Telangana) for Mango, Anantpur (A.P.) and Theni (T.N.) for Banana, Nasik (Maharashtra) for Grapes, Siphahijala (Tripura) for Pineapple, Solapur (Maharashtra) and Chitradurga (Karnataka) for Pomegranate and West Jaintia Hills (Meghalaya) for Turmeric. These clusters will be implemented through Cluster Development Agencies (CDAs) which are appointed on the recommendations of the respective State/UT Government.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1723155>

Q.21) Ans: B

Exp:

Millets

- Millets are a group of highly variable small-seeded grasses, widely grown around the world as cereal crops or grains for fodder and human food. Most species generally referred to as millets belong to the tribe Paniceae, but some millets also belong to various other taxa.
- Bajra (pearl millet), jowar (sorghum), ragi (finger millet), kodo (kodo millet), kutki (little millet), kakun (foxtail millet), sanwa (barnyard millet), cheena (proso millet), kuttu (buckwheat) and chaulai (amaranth) are high-nutrient different types of millets.
- The Union Agriculture Ministry, in April 2018, declared millets as "Nutri-Cereals", considering their "high nutritive value" and also "anti-diabetic properties".
- The year 2018 was observed as the "National Year of Millets". The UN General Assembly adopted an India-sponsored resolution to mark 2023 as the "International Year of Millets".

Source: <https://indianexpress.com/article/india/in-madhya-pradeshs-dindoi-a-un-backed-project-aims-to-put-mil>

Q.22) Ans: D

Exp:

Entomopathogenic fungi

- An entomopathogenic fungus is a fungus that can act as a parasite of insects and kills or seriously disables them. Entomopathogenic fungi (EPF) are

common in terrestrial environments and can be important natural regulators of insect and arachnid populations.

- Entomopathogenic fungi, unlike other groups of insect pathogenic microorganisms, infect their hosts directly through the exoskeleton. In contrast, insect-associated viruses, bacteria and microsporidia penetrate and infect the host via the mid-gut following ingestion.
- Entomopathogenic fungi specifically infect and often kill insects and other arthropods. Most are non pathogenic to plants, and relatively non-toxic to humans and animals. Though fungus-infected insects can be commonly found in nature, and epizootics are observed impacting pest populations, mortality from fungal infection rarely occurs naturally at sufficiently high levels or early enough in a pest cycle to prevent economic loss.

Source: <https://www.thehindu.com/sci-tech/science/how-whiteflies-came-saw-and-conquered-indias-crops/article34623516.ece> and <https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/entomopathogenic-fungi>

Q.23) Ans: C

Exp:

'Harit Dhara'

- Indian Council of Agricultural Research (ICAR) institute has developed an anti-methanogenic feed supplement 'Harit Dhara'. When given to bovines and sheep, it not only cuts down their methane emissions by 17-20%, but also results in higher milk production and body weight gain. In other words, win-win for both the environment and livestock farmers.
- Harit Dhara acts by decreasing the population of protozoa microbes in the rumen, responsible for hydrogen production and making it available to the archaea for reduction of CO₂ to methane.
- The Indian Council of Agricultural Research (ICAR) is an autonomous body responsible for co-ordinating agricultural education and research in India. It reports to the Department of Agricultural Research and Education, Ministry of Agriculture. The Union Minister of Agriculture serves as its president.

Source: <https://indianexpress.com/article/india/feed-developed-by-icar-also-increases-weight-milk-production-feed-supplement-reduces-methane-emissions-by-livestock-7389140/>



Q.24) Ans: C

Exp:

- Ranjit Sub1, Swarna Sub1 and Bahadur Sub1 are flood-tolerant paddy varieties developed by the Indian Council of Agricultural Research and the Manila-based International Rice Research Institute.
- The 'sub' mentioned in the varieties is the introduced submergence genes which are responsible for greater yields under flooded conditions.
- With its major rivers the Brahmaputra and Barak, which have more than 50 tributaries, the state of Assam is prone to floods and erosion. Low-lying areas remain submerged for as long as two weeks after flash floods, so switchover from traditional varieties of paddy is of prime importance.

Source:

<https://www.thehindu.com/news/national/other-states/amid-deluge-assam-farmer-stands-out-with-flood-resistant-paddy/article32060811.ece>

Q.25) Ans: C

Exp:

- **Statement 1 is correct** : Biotech-Krishi Innovation Science Application Network (Biotech-KISAN) is a **Department of Biotechnology, Ministry of Science and Technology** initiative that empowers farmers, especially women farmers. It aims to understand the problems of water, soil, seed and market faced by the farmers and provide simple solutions to them. The Scheme is for farmers, developed by and with farmers, it empowers women, impacts locally, connects globally, is Pan-India, has a hub-and-spoke model and stimulates entrepreneurship and innovation in farmers.
- **Statement 2 is correct** : Biotech-Krishi Innovation Science Application Network (Biotech-KISAN) is **being implemented in 15 agro-climatic zones of India in a phased manner** with the following objectives:
 - Linking available science and technology to the farm by first understanding the problem of the local farmer and providing solutions to those problems.
 - The working together, in close conjunction, of scientists and farmers is the only way to improve the working conditions of small and marginal farmers.
 - This programme aims to work with small and marginal farmers, especially the

woman farmer, for better agricultural productivity through scientific intervention and evolving best farming practices in the Indian context.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1728507>

Q.26) Ans: D

Exp:

- **Statement 1 is incorrect** : The plant was developed at the Kerala-based Rubber Research Institute of India (RRII) and it planted the world's first Genetically Modified (GM) rubber plant on the outskirts of Guwahati in Assam. The GM rubber has additional copies of the gene MnSOD, or manganese-containing superoxide dismutase, inserted in the plant, which is expected to tide over the severe cold conditions during winter — a major factor affecting the growth of young rubber plants in the region.
- **Statement 2 is incorrect** : As per the Ministry of Commerce and Industry Kerala is the biggest producer of natural rubber. Natural rubber is cultivated in 16 states in India. With over 600,000 hectares, Kerala tops rubber cultivation, followed by Tripura with over 85,038 hectares under plantation. Other major natural rubber producers are Karnataka, Tamil Nadu, Assam and other northeastern states.

Source:

<https://www.thehindu.com/news/national/other-states/first-ever-genetically-modified-rubber-planted-in-assam/article34901294.ece>

Q.27) Ans: D

Exp:

- **Statement 1 is incorrect**: In the budget speech of 2018-19, a new Scheme "Operation Greens" was announced on the line of "Operation Flood". **It seeks to stabilize the supply of Tomato, Onion and Potato (TOP) crops and to ensure availability of TOP crops throughout the country round the year without price volatility.** The scheme was extended during June 2020 to cover all fruits & vegetables (TOTAL) for a period of six months on pilot basis as part of Aatmanirbhar Bharat Abhiyan. The Government of India, in its Union budget 2021, announced that Operation Green (OG) will be

expanded beyond tomatoes, onions, and potatoes (TOP) to 22 perishable commodities.

- **Statement 2 is incorrect:** The Ministry of Food Processing Industries has launched the scheme. NAFED will be the Nodal Agency to implement price stabilisation measures. **Objectives of schemes -**
 - Enhancing value realisation of TOP farmers by targeted interventions to strengthen TOP production clusters and their FPOs, and linking/connecting them with the market.
 - Price stabilisation for producers and consumers by proper production planning in the TOP clusters and introduction of dual use varieties.
 - Reduction in post-harvest losses by creation of farm gate infrastructure, development of suitable agro-logistics, creation of appropriate storage capacity linking consumption centres.
 - Increase in food processing capacities and value addition in TOP value chain with firm linkages with production clusters.
 - Setting up of a market intelligence network to collect and collate real time data on demand and supply and price of TOP crops.

Source:

<https://indianexpress.com/article/opinion/columns/nirmala-sitharaman-union-budget-milk-production-dairy-products-7208925/>

Q.28) Ans: D

Exp:

- Technical Assistance for Development of Sustainable Agricultural Value Chains in Mizoram:- Project is implemented in the State of Mizoram to strengthen capacities of government institutions and extension services in Mizoram to support local communities in improved jhum management, sustainable value chain development, and exposure of project staff to best agro-ecological practices.

Source:

<https://agricoop.nic.in/sites/default/files/Web%20copy%20of%20AR%20%28Eng%29.pdf>

Page: 201

Q.29) Ans: A

Exp:

- **Agricultural yield** is a standard measurement of the amount of agricultural production harvested—yield of a crop—per unit of land area. It is measured for cereal, grain, or legumes; and typically is measured in tons, per hectare.
- India's agricultural yield is found to be lower in the case of most crops, as compared to other top producing countries such as China, Brazil and the United States.
- **Yield potential** is defined as the yield of a cultivar when grown in environments to which it is adapted, with nutrients and water non-limiting and with pests, diseases, weeds, lodging, and other stresses effectively controlled.
- **'Crop yield'** is the weight of grain or other economic product, at some agreed standard moisture content, per unit of land area harvested per crop (usually metric tons per hectare¹, or here abbreviated to t/ha).
- **Agricultural productivity** is measured in money produced per unit of land, but yields are measured in the weight of the crop produced per unit of land.

Source: <https://prsindia.org/policy/analytical-reports/state-agriculture-india>

Q.30) Ans: C

Exp:

Fogponics

- Fogponics is a sub-technique of aeroponics. Unlike traditional Hydroponics, where plants get the oxygen from the water oxygenated by an air pump, Fogponics/Aeroponics works **by hanging the plants in the air**. It is held by a tray/foam. The plant roots are suspended, so they have plenty of oxygen access. Water is provided by plants via the foggers
- The fogponics system uses foggers, which pump and vibrate under pressure to transform a nutrient and water mixture into humidity, not dissimilar to the humid atmosphere one might find in the rainforest. This fog creates a constant nutrient-rich environment for plant roots.
- Nutrients are delivered via foggers which usually run on a timer. The benefit of using foggers rather than sprayers or misters like in traditional aeroponics is that foggers can atomize water droplets into even smaller particles, usually less than 10 microns in size.

Source:

<https://agricoop.nic.in/sites/default/files/Web%20copy%20of%20AR%20%28Eng%29.pdf>



Q.31) Ans: C

Exp:

- **Statement 1 is correct:** Central Research Institute for Dryland Agriculture (CRIDA) is a National Research Institute under the Indian Council of Agricultural Research (ICAR) established in 1985 with a mandate to carry out basic and applied research in rainfed farming. The Institute also undertakes National / International Collaborations and Consultancy Projects. All India Coordinated Research Programmes (AICRPs) of ICAR on Dryland Agriculture and Agrometeorology with 25 partners each are in CRIDA. This is the lead Institute and the National Nodal point for the National Innovations in Climate Resilient Agriculture (NICRA) which is being implemented at a large number of Research Institutes of ICAR, State Agricultural Universities and 100 KVKs.
- **Statement 2 is correct :** Central Research Institute for Dryland Agriculture (CRIDA) has developed detailed district-wise contingency plans to provide a broad advisory to farmers. These contingency plans prescribe alternate strategies in the event of climate variability by factoring in crops/livestock/aquaculture practices/pattern, soil characteristics, infrastructural facilities, etc. These plans have been developed based on certain simulated models for different weather conditions like occurrences of flood, drought, cyclone, cold wave/frost, etc. CRIDA has so far prepared Contingency Plans for 650 districts of the country.

Source: <http://www.icar-crida.res.in/>

Q.32) Ans: C

Exp:

- The **Ministry of Agriculture & Farmers Welfare** launched a **Global Environment Facility (GEF)** assisted project namely, “Green – Ag: Transforming Indian Agriculture for global environment benefits and the conservation of critical biodiversity and forest landscapes” in collaboration with the **Food and Agriculture Organisation (FAO)** in high-conservation-value landscapes.
- The overall objective of the project is to catalyze **transformative change of India’s agricultural sector to support achievement of national and global environmental benefits and conservation of critical biodiversity and forest landscapes.**
- Landscapes of five States are identified under this project, namely:

- (i) Madhya Pradesh : Chambal Landscape,
- (ii) Mizoram: Dampa Landscape,
- (iii) Odisha: Similipal Landscape,
- (iv) Rajasthan: Desert National Park Landscape and
- v) Uttarakhand: Corbett-Rajaji Landscape.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1558249>

Q.33) Ans: C

Exp:

- The **Ministry of Agriculture & Farmers Welfare** launched a **Global Environment Facility (GEF)** assisted project namely, “Green – Ag: Transforming Indian Agriculture for global environment benefits and the conservation of critical biodiversity and forest landscapes” in collaboration with the **Food and Agriculture Organisation (FAO)** in high-conservation-value landscapes.
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 - (iii) Odisha: Similipal Landscape,
 - (iv) Rajasthan: Desert National Park Landscape and
 - v) Uttarakhand: Corbett-Rajaji Landscape.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1558249>

Q.34) Ans: C

Exp:

The National Green Tribunal resolves various civil cases under the following **seven laws** that are related to the environment:

- Water Act (Prevention and Control of Pollution), 1974
- Water Cess Act (Prevention and Control of Pollution), 1977
- Forest Act (Conservation), 1980



- Air Act (Prevention and Control of Pollution) 1981
- Environment (Protection) Act, 1986
- Public Liability Insurance Act, 1991
- Biological Diversity Act, 2002

The following acts have been kept out of the purview of NGT:

- The Indian Forest Act, 1927
- The Wildlife (Protection) Act, 1972
- Forest Rights Act, 2006

NGT:

- It is a specialised body set up under the National Green Tribunal Act (2010) for effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources.
- With the establishment of the NGT, India became the third country in the world to set up a specialised environmental tribunal, only after Australia and New Zealand, and the first developing country to do so.
- NGT is mandated to dispose of applications or appeals within 6 months of filing of the same.
- The NGT has five places of sittings, New Delhi is the Principal place of sitting and Bhopal, Pune, Kolkata and Chennai are the other four.

Q.35) Ans: C

Exp:

- **Statement 1 is correct:** The genesis of the Environmental (Protection) Act, 1986, is in Article 48A (Directive Principles of State Policy) and Article 51A (g) (Fundamental Duties) of the Indian Constitution.
- **Statement 2 is incorrect:** The Environment (Protection) Act, 1986 has relaxed the rule of "Locus Standi" and because of such relaxation, even a common citizen can approach the Court provided he has given a notice of sixty days of the alleged offense and his intention to make a complaint to the Central Government or any other competent authority.
- **Statement 3 is correct:** It grants immunity to the officers of the Government for any act done under the provisions of this Act or the powers vested in them or functions assigned to them under this Act.
- **Statement 4 is incorrect:** The Act debars the Civil Courts from having any jurisdiction to entertain any suit or proceeding in respect of an action,

direction, order issued by the Central Government, or other statutory authority under this Act.

Q.36) Ans: C

Exp:

- The Central Pollution Control Board (CPCB) is a statutory organization constituted under the **Water (Prevention and Control of Pollution) Act, 1974.**
- Central Pollution Control Board and the State Boards to encourage conservation of water. Accordingly, the State Pollution Control Board are collecting water cess from the categories of industries as mentioned in the "SCHEDULE I" of the **Water (Prevention and Control of Pollution) Cess Act, 1977** and as amended upto date.
- It was entrusted with the powers and functions under the **Air (Prevention and Control of Pollution) Act, 1981.**
- It serves as a field formation and also provides technical services to the Ministry of Environment, Forests and Climate Change (MoEF&CC) of the provisions of the **Environment (Protection) Act, 1986.**
- **The Forest Conservation Act, 1980** was passed to check further deforestation and conserve forests.
- **Forest Rights Act, 2006** provides for the restitution of deprived forest rights across India, including both individual rights to cultivated land in forestland and community rights over common property resources.

Q.37) Ans: D

Exp:

- The **NAAQS** have been revisited and revised in November 2009 for 12 pollutants, which include- sulphur dioxide (SO₂), nitrogen dioxide (NO₂), particulate matter having size less than 10 micron (**PM 10**), particulate matter having size less than 2.5 micron (PM_{2.5}), ozone, **lead**, carbon monoxide (**CO**), arsenic, nickel, benzene, **ammonia** (NH₃), and Benzopyrene.
- National Air Quality Index (**AQI**) considers eight pollutants: **PM 10**, PM 2.5, Ozone (O₃), Sulphur dioxide (SO₂), nitrogen dioxide (NO₂), carbon monoxide (**CO**), **lead** (Pb) and **ammonia** (NH₃).



Q.38) Ans: D

Exp:

- **Statement 1 is not correct:** Densely populated rural areas to be afforded greater opportunity for development: For CRZ-III (Rural) areas, two separate categories have now been stipulated as below:
- **CRZ-III A** - These are densely populated rural areas with a population density of 2161 per square kilometer as per the 2011 Census. Such areas shall have a No Development Zone (NDZ) of 50 meters from the HTL as against 200 meters from the High Tide Line stipulated in the CRZ Notification, 2011 since such areas have similar characteristics as urban areas.
- **CRZ-III B** - Rural areas with a population density of below 2161 per square kilometer as per the 2011 Census. Such areas shall continue to have an NDZ of 200 meters from the HTL.
- **Statement 2 is not correct:** Temporary tourism facilities such as shacks, toilet blocks, change rooms, drinking water facilities, etc. have now been permitted in Beaches. Such temporary tourism facilities are also now permissible in the "No Development Zone" (NDZ) of the CRZ-III areas as per the Notification. However, a minimum distance of 10 m from HTL should be maintained for setting up of such facilities.
- **Statement 3 is not correct:** As per CRZ, 2011 Notification, for CRZ-II (Urban) areas, Floor Space Index (FSI), or the Floor Area Ratio (FAR) had been frozen as per 1991 Development Control Regulation (DCR) levels. In the CRZ, 2018 Notification, it has been decided to **de-freeze** the same and permit FSI for construction projects.

Q.39) Ans: C

Exp:

- **Statement 1 is correct:** The Environment (Protection) Act was enacted in 1986 with the objective of providing for the protection and improvement of the environment. It empowers the Central Government to establish authorities [under section 3(3)] charged with the mandate of preventing environmental pollution in all its forms and to tackle specific environmental problems that are peculiar to different parts of the country.

As per the EPA "hazardous substances" means any substance which, by reason of its chemical or physio-

chemical properties or handling, is liable to cause harm to human beings and other living creatures, plants, micro organisms, property or environment.

- **Statement 2 is correct :** The Central Government shall have the power to take all such measures regarding prevention, control and abatement of environmental pollution and as it deems necessary or expedient for the purpose of protecting and improving the quality of the environment in coordination with the State Governments but no such formulation and implementation powers are provided to the state governments. It is considered as one of the potential drawbacks of the EPA.

Source:

<https://cpcb.nic.in/displaypdf.php?id=aG9tZS9lcGEvZXByb3RIY3RfYWNOXzE5ODYucGRm>

Q.40) Ans: D

Exp:

E-waste (Management) Amendment Rules, 2018

- The Union Government has amended the E-waste (Management) Rules in a move to facilitate and effectively implement the environmentally sound management of e-waste in India.
- The amendment in rules has been done with the objective of channelizing the E-waste generated in the country towards authorized dismantlers and recyclers in order to formalize the e-waste recycling sector.
- Some of the **salient features of the E-waste (Management) Amendment Rules, 2018** are as follows:
 - The e-waste collection targets under EPR have been revised and will be applicable from 1 October 2017.
 - The phase-wise collection targets for e-waste in weight shall be 10% of the quantity of waste generation as indicated in the EPR Plan during 2017-18, with a 10% increase every year until 2023. After 2023 onwards, the target has been 70% of the quantity of waste generated as indicated in the EPR Plan.
 - **Separate e-waste collection targets** have been drafted for new producers, i.e. those producers whose number of years of sales operation is less than the average life of their products. The average life of the products will be as per the guidelines issued by CPCB from time to time.

- **Producer Responsibility Organizations** (PROs) shall apply to the Central Pollution Control Board (CPCB) for registration to undertake activities prescribed in the Rules.
- Under the Reduction of Hazardous Substances (RoHS) provisions, cost for sampling and testing shall be borne by the government for conducting the RoHS test. If the product does not comply with RoHS provisions, then the cost of the test will be borne by the Producers.

Q.41) Ans: A

Exp:

- **Section 2(b) of the Scheduled Tribes And Other Traditional Forest Dwellers (Recognition Of Forest Rights) Act, 2006** defines “critical wildlife habitat” as such areas of **National Parks and Sanctuaries** where it has been specifically and clearly established, case by case, **on the basis of scientific and objective criteria**, that such areas are required to be kept as **inviolable for the purposes of wildlife conservation**.
- Although the **Ministry of Tribal Affairs** is the **nodal ministry for the Forest Rights Act, 2006**, the CWHs are determined and notified by the **Central Government in the Ministry of Environment and Forests**.

SOURCE: Critical Wildlife Habitat guidelines issued; NTCA order superseded (downtoearth.org.in)

Q.42) Ans: B

Exp:

- **Statement 1 is incorrect:** Deemed forests, comprising about 1% of India’s forest land, are a controversial subject as they refer to land tracts that appear to be a “forest”, but have not been notified so by the government or in historical records. **The concept of deemed forests has not even been clearly defined in any law including the Forest Conservation Act 1980.** The definition of deemed forests varies from state to state.
 - In the **T N Godavarman Thirumulpad case 1996**, the Supreme Court (SC) accepted a wide definition of forests under the Act and held that the **word ‘forest’ must be understood according to its dictionary meaning**.
- **Statement 2 is correct:** According to **Section 2 of the act**, , no **State Government** or other authority

shall make, **except with the prior approval of the Central Government**, any order directing-

- that any reserved forest shall cease to be reserved.
- that **any forest land or any portion thereof may be used for any non-forest purposes**.
- that any forest land or any portion thereof may be cleared of trees which have grown naturally in that land or portion, for the purpose of using it for afforestation.
- that any forest land or any portion thereof may be assigned by way of lease or otherwise to any private person or to any authority, corporation, agency or any other Organisation not owned, managed or controlled by Government.

SOURCES : Forest Conservation Act 1980 (nbaindia.org)

Union Environment Ministry to challenge Uttarakhand forest notice - The Hindu
Explained: What are deemed forests, and why Karnataka wants to declassify some forests

Q.43) Ans: B

Exp:

- Recently, the Ministry for Environment, Forest and Climate Change (MoEF&CC) has released **Management Effectiveness Evaluation (MEE) of 146 national parks and wildlife sanctuaries (NP&WLS)** in the country.
- Management Effectiveness Evaluation (MEE) is the assessment of how well protected areas such as **national parks, wildlife sanctuaries, conservation reserves, community reserves and tiger reserves** are being managed and their effectiveness in conserving target flora and fauna.
 - The assessment process of India’s National Park and Wildlife Sanctuaries adopted from IUCN WCPA (World Commission on Protected Areas) framework of MEE.
 - The MoEF&CC has also launched the **Management Effectiveness Evaluation of Indian Zoos (MEE-ZOO) framework** which proposes guidelines, criteria and indicators for evaluation of zoos of the country in a manner which is discrete, holistic and independent.

SOURCE: [mee-writeup.pdf \(cbd.int\)](#)

Q.44) Ans: B

Exp:

- Options 1,4 and 5 are correct:
 - In exercise of the powers conferred by the **Environment (Protection) Act, 1986**, the **Central Government** has published the **Wetlands (Conservation and Management) Rules, 2017** according to which wetland means
 - an **area of marsh, fen, peatland or water;**
 - whether **natural or artificial, permanent or temporary**, with water that is **static or flowing, fresh, brackish or salt**, including areas of marine water the depth of which at low tide does not exceed six meters
- The definition however **does not include**
 - river channels
 - **paddy fields**
 - human-made water bodies/tanks specifically constructed for drinking water purposes
 - structures specifically constructed for **aquaculture, salt production, recreation and irrigation purposes.**

SOURCE: Wetlands Conservation Management Rules 2017 (yamuna-revival.nic.in)

Q.45) Ans: C

Exp:

- Option C is correct:
 - The **Wildlife (Protection) Act, 1972** was enacted for the protection of plants and animal species. It prohibits the **uprooting, damage, collection, possession or selling of any specified plant from any forest land or any protected area.**
 - **Exception:** The CWLW, however, may grant permission for uprooting or collecting a specific plant for the **purpose of education, scientific research, preservation in a herbarium** or if a person/institution is approved to do so by the central government.
- However, The Act has divided the protection status of various plants and animals under 6 schedules. The **VIth schedule provides for regulation in cultivation of a specified plant and restricts its possession, sale and transportation.**

- Both **cultivation and trade of specified plants can only be carried out with prior permission of competent authority.**
- Plants protected under **Schedule VI** include:
 - **Beddome's cycad (Native to India)**
 - **Blue Vanda (Blue Orchid)**
 - **Red Vanda (Red Orchid)**
 - **Kuth (Saussurea lappa)**
 - **Slipper orchids (Paphiopedilum spp.)**
 - **Pitcher plant (Nepenthes khasiana)**

SOURCE: A1972-53_0.pdf (legislative.gov.in)

Q.46) Ans: A

Exp:

- Options 1 and 4 are correct:
 - According to the **Biological Diversity Act, 2002**, **Biological resources** means:
 - **plants, animals and microorganisms or parts thereof,**
 - Their **genetic material and by-products (excluding value added products)** with actual or potential use or value
 - Here, **"value added products"** means products which may contain portions or extracts of **plants and animals in unrecognizable and physically inseparable form**
- However, the definition of biological resources **does not include genetic material.**
- In addition to this, **"biological diversity"** means the variability among living organisms from **all sources and the ecological complexes** of which they are part and includes diversity **within species or between species and of ecosystems.**

SOURCE:

<https://www.indiacode.nic.in/bitstream/123456789/2046/1/200318.pdf>

Q.47) Ans: B

Exp:

- **Maharashtra** became the first state in the country to adopt the Fly Ash Utilization Policy. The policy paved the way for prosperity by generating wealth from waste and environment protection.
- The policy seeks 100% use of fly ash generated from thermal power plants and biogas plants for



construction activities. Facilitates use of fly ash to make bricks, blocks, tiles, wall panels, cement and other construction materials. The policy extends use of fly ash to 300 kms radius of power plant from earlier 100 kms radius of power plant.

- It will help in environment protection and save soil excavation. It also will make available raw material for construction at low cost to help 'Housing for All' projects.
- It will also help create new employment opportunities in the power plant sector and pave the way for prosperity by generating Wealth from Waste.

• **Key Facts About Fly Ash**

- Fly ash is a fine, glass powder by-product recovered from gases of burning coal in thermal power plants during production of electricity. They are micron sized earth elements primarily consisting of silica, alumina and iron.
- The Fly ash causes air pollution. It can also contaminate water and soil systems. The wet disposal of Fly ash results in leaching of toxic heavy metals in groundwater systems.
- Fly ash can be used as a replacement for some of the Portland cement contents of concrete. It can be used in the production of bricks for building construction.
- The Central Government has made it mandatory for use of fly ash bricks in construction activities happening 500km around thermal power plants.

receiving boosts from Earth's gravity, Lucy will embark on a 12-year journey to eight different asteroids, one in the Main Belt between Mars and Jupiter and the seven TrojAns: Lucy will fly by its target objects within 250 miles (400 kilometers) of their surfaces, and use its onboard instruments and large antenna to investigate their geology, including composition, mass, density and volume.

Source - <https://www.thehindu.com/sci-tech/science/nasas-lucy-mission-to-probe-jupiters-mysterious-trojan-asteroids/article36747491.ece/amp/>

Q.49) Ans: B

Exp:

- **Statement 1 is incorrect** : Rashtriya Mahila Kosh was established in 1993 as a national level autonomous body under the Ministry of Women & Child Development for socio-economic empowerment of women, especially those in the rural and unorganized sector, by providing them concessional, collateral free micro-credit. It was registered under the Societies Registration Act 1860. Recently, the GOI has decided to close down the credit facilitating agency Rashtriya Mahila Kosh as it has lost relevance with alternative credit facilities becoming available to women through various government initiatives and efforts such as Jan Dhan Yojana, PM Mudra Yojana etc and also to avoid duplicacy of efforts.
- **Statement 2 is correct** : RMK extends concessional microfinance loans to poor women through Intermediary Organizations for livelihood activities, housing, micro-enterprises and family needs to bring about socio-economic upliftment of the women. The operating model currently followed by RMK is a group model with RMK as a facilitating agency wherein RMK provides loan products to Non – Governmental Organization (NGO)/ Intermediary Micro-financing Organization (IMO) / Voluntary Organisation (VO) which on-lend to Women Groups like SHG, JLG etc

Source -

https://m.economictimes.com/news/economy/policy/rashtriya-mahila-kosh-lost-relevance-is-being-closed-govt/amp_articleshow/85070012.cms

Q.50) Ans: A

Exp:

Q.48) Ans: C

Exp:

- **Statement 1 is correct** : NASA is poised to send its first spacecraft to study Jupiter's Trojan asteroids to glean new insights into the solar system's formation 4.5 billion years ago. The probe, called Lucy after an ancient fossil that provided insights into the evolution of human species, will launch on October 16 from Cape Canaveral Space Force Station in Florida. It will be the **first solar-powered to venture** this far from the Sun, and will observe more asteroids than any other spacecraft before it.
- **Statement 2 is correct** : Its mission is to investigate the group of rocky bodies circling the Sun in two swarms, one preceding Jupiter in its orbital path and the other trailing behind it. After

- **Statement 1 is Correct: To phase out single use plastic by 2022**, keeping in view the adverse impacts of littered plastic on both terrestrial and aquatic ecosystems, the **Ministry of Environment, Forest and Climate Change** has notified the **Plastic Waste Management Amendment Rules, 2021**, which prohibits identified single use plastic items which have low utility and high littering potential by 2022.
- **Statement 2 is correct: Extended Producer Responsibility (EPR)** is a policy approach under which producers are given a significant responsibility – financial and/or physical – for the treatment or disposal of post-consumer products. The plastic packaging waste, which is not covered under the phase out of identified single use plastic items, shall be collected and managed in an environmentally sustainable way through the Extended Producer Responsibility of the **Producer, importer and Brand owner (PIBO)**, as per Plastic Waste Management Rules, 2016. For effective implementation of Extended Producer Responsibility the Guidelines for Extended Producer Responsibility being brought out have been given legal force through Plastic Waste Management Amendment Rules, 2021.
- **Statement 3 is incorrect:** In order to stop littering due to light weight plastic carry bags, with effect from **30th September, 2021, the thickness of plastic carry bags has been increased from 50 microns to 75 microns and to 120 microns with effect from the 31st December, 2022**. This will also allow reuse of plastic carry due to increase in thickness.

Extra edge by OnlyIAS :

- According to Rules, 2021 thickness of plastic carry bags increased from 50 to 75 microns from 30th September, 2021 and 120 microns with effect from the 31st December, 2022.

Source: PIB- 13th August, 2021. pib.gov.in

Q.51) Ans: B

Exp:

- **Statement 1 is correct:** The “Ubharte Sitaare Fund” is set up to enhance India’s export competitiveness and to identify Indian enterprises with potential advantages by way of technology, products or processes along with

export potential, but which are currently underperforming or unable to tap their latent potential to grow. The fund is a mix of structured support, both financial and advisory services through investments in equity or equity like instruments, debt (funded and non-funded) and technical assistance (advisory services, grants and soft loans) to the Indian companies.

- **Statement 2 is incorrect:** The 'Ubharte Sitaare Fund' has been set up by Exim Bank and SIDBI, Not by RBI.

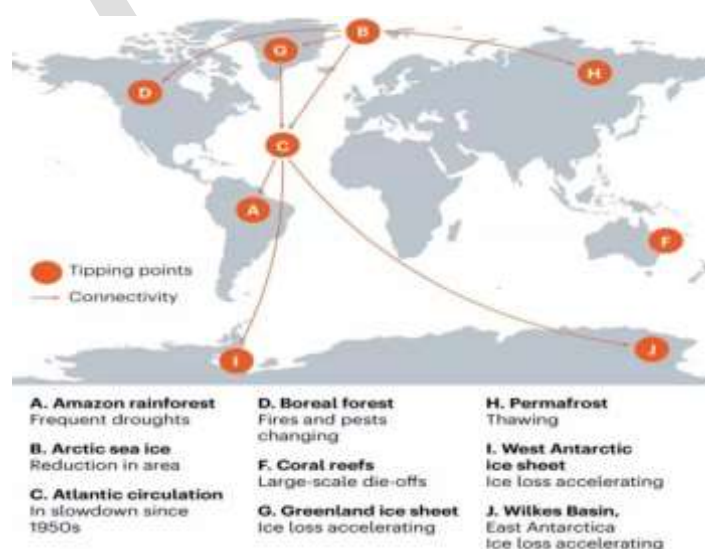
Source- The Hindu thehindu.com

Q.52) Ans: C

Exp:

Climate Tipping Points:

- Recently it has been in the news because of the sixth assessment report of the IPCC it is very likely that AMOC will decline over the 21st century. And, AMOC is one of the nine tipping points where a changing climate could push parts of the earth system into abrupt or irreversible change.
- Tipping Points- “These are thresholds where a tiny change could push a system into a completely new state”.
- Globally there are 9 tipping points where a changing climate could push parts of the earth system into abrupt or Irreversible change.



Source- Down to Earth.

Q.53) Ans: A

Exp:



About e-RUPI

- e-RUPI is a digital voucher which a beneficiary gets on his phone in the form of an SMS or QR code. It is a pre-paid voucher, which can be redeemed at any centre that accepts it.
- e-RUPI has launched by the National Payments Corporation of India (NPCI), which oversees the digital payments ecosystem in India.
- NPCI has partnered with 11 banks for e-RUPI transactions. They are Axis Bank, Bank of Baroda, Canara Bank, HDFC Bank, ICICI Bank, Indian Bank, IndusInd Bank, Kotak Mahindra Bank, Punjab National Bank, State Bank of India and Union Bank of India.

Advantages of e-RUPI:

- e-RUPI does not require the beneficiary to have a bank account, a major distinguishing feature as compared to other digital payment forms.
- Thus e-RUPI is a one time contactless, cashless voucher-based mode of payment that helps users redeem the voucher without a card, digital payments app, or internet banking access.
- It ensures an easy, contactless two-step redemption process that does not require sharing of personal details either.
- e-RUPI is operable on basic phones also, and hence it can be used by persons who do not own smart-phones or in places that lack internet connection.
- Being a prepaid voucher, e-RUPI would assure real time payments to the service provider.

Q.54) Ans: D

Exp:

- **Statement 1 is incorrect:** It is a Maritime exercise not a military exercise, for shared vision for a free, open and inclusive Indo-Pacific region of participating navies.
- **Statement 2 is incorrect:** Malabar series of maritime exercises commenced in 1992 as an IN-USN Exercise. In 2015, JMSDF (Japan) joined Malabar as a permanent member. The 2020 edition witnessed participation of the Royal Australian Navy.
 - MALABAR-21 would witness complex exercises including anti-surface, anti-air and anti-submarine warfare drill, and other manoeuvres and tactical exercises. The exercise will provide an opportunity

for participating navies to derive benefit from each other's expertise and experiences.

Source- pib pib.gov.in

Q.55) Ans: B

Exp:

- **Statement 1 is correct:** The Deep Ocean Mission aims to explore the deep ocean for resources. Also, for sustainable use of ocean resources, it aims to develop deep-sea technologies. The Ministry of Earth Sciences (MoES) will be the nodal Ministry for implementing this mission. The mission will be implemented over a period of 5 years in a phased manner. Its 1st phase will be of 3 years.
- **Statement 2 is correct:** Key Components of the mission
 - An **Integrated Mining System** will be developed for mining polymetallic nodules at those depths in the central Indian Ocean.
 - **Development of Ocean Climate Change Advisory Services.** It helps in searching for deep sea flora and fauna, including microbes, and studying ways to sustainably utilise them.
 - **To explore and identify potential sources of hydrothermal minerals** that are sources of precious metals formed from the earth's crust along the Indian Ocean mid-oceanic ridges.
 - **Studying and preparing detailed engineering design for offshore Ocean Thermal Energy Conversion (OTEC) powered desalination plants.**
 - Grooming experts in the field of ocean biology and engineering. This component aims to translate research into industrial applications and product development through on-site business incubator facilities.
- **Statement 3 is incorrect:** The mission will give a boost to efforts to explore India's vast Exclusive Economic Zone and Continental Shelf. The Mission aims to support India's Blue Economy Initiatives. The Ministry of Earth Sciences has a 15-year contract with the International Seabed Authority (ISA), an UN-backed body in charge of regulations for ocean floors, to explore 75,000 sq. km in the Indian Ocean for Manganese Nodules, also known as Polymetallic Nodules. The plan will enable India



- to develop capabilities to exploit resources in the Central Indian Ocean Basin (CIOB).

<https://pib.gov.in/PressReleasePage.aspx?PRID=1727525>

Extra Edge by Only IAS

- The United Nations(UN) has declared the decade 2021-2030 as the Decade of Ocean Science for Sustainable Development. The Government of India's Vision of New India by 2030 announced in 2019 highlighted the Blue Economy as one of the ten core dimensions of growth.
- Polymetallic nodules** : Polymetallic nodules (also known as manganese nodules) are potato-shaped, largely porous nodules found in abundance carpeting the sea floor of world oceans in deep sea.
- Composition** : Besides manganese and iron, they contain nickel, copper, cobalt, lead, molybdenum, cadmium, vanadium, titanium, of which nickel, cobalt and copper are considered to be of economic and strategic importance.

Q.56) Ans: A

Exp:

- Statement 1 is correct:** Industrial deep Decarbonisation is a Clean Energy Ministerial's (CEM's) initiative. It is a global coalition of public and private organisations who are working to stimulate demand for low carbon industrial materials. In collaboration with national governments, IDDI works to standardise carbon assessments, establish ambitious public and private sector procurement targets, incentivise investment into low-carbon product development and design industry guidelines.
- Statement 2 is correct:** Industrial deep Decarbonisation initiative is Coordinated by the UN Industrial Development Organization (UNIDO), the IDDI is co-led by the UK and India and current members include Germany and Canada. The initiative brings together a strong coalition of related initiatives and organizations including the Mission Possible Platform, the Leadership Group for the Industry Transition, the International Renewable Energy Agency (IRENA) and the World Bank to tackle carbon intensive construction materials such as steel and cement.

<https://www.unido.org/IDDI>

Extra Edge by Only IAS

Clean Energy Ministerial

- It was established in December 2009 at the UN's Framework Convention on Climate Change conference of parties in Copenhagen.
- Purpose of CEM is a high-level global forum to promote policies and programs that advance clean energy technology, to share lessons learned and best practices, and to encourage the transition to a global clean energy economy.
- 29 countries are part of CEM and India is also a member country.

United Nations Development Industrial Organisation

- UNIDO is the specialized agency of the United Nations that promotes industrial development for poverty reduction, inclusive globalization and environmental sustainability.
- There are 170 Member countries as of 1st April 2019. India is also a member country.
- UNIDO's mandate is fully recognized in SDG-9, which calls to "Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation".
- It's headquarters is in Vienna, Austria.

Q.57) Ans: C

Exp:

- Statement 1 is incorrect:** Equalisation Levy was chargeable only on the consideration received or receivable by a non-resident from providing online advertisement services or related services to the Indian resident or person having PE in India which are liable to pay Equalisation levy @6%. New Amendment to Finance Act, 2020 widen the scope of Equalisation Levy by including within its ambit the consideration received or receivable for E-commerce Supply or Services by an e-commerce operator. Now, e-commerce operators will be liable to pay Equalisation Levy @2%.
- Statement 2 is incorrect:** It is introduced by Amendment to Finance Act, 2020 by taxing foreign e-commerce sellers to level the playing field with local businesses who pay taxes in India. The tax applies on e-commerce transactions on websites such as Amazon.com. Google in particular as the tax applies on advertising revenue earned overseas if those ads target customers in India. Offshore e-commerce firms that have a permanent establishment in India or pay income tax in India are exempt from the two percent equalization levy.



<https://www.india-briefing.com/news/indias-digital-tax-2-percent-not-applicable-foreign-e-commerce-companies-indian-arm-21956.html/>

Q.58) Ans: A

Exp:

- **Statement 1 is correct:** NHRC, established in 1993, is an independent statutory body as per the provisions of the Protection of Human Rights Act of 1993. Its objective is to protect and promote human rights related to life, liberty, equality and dignity of individuals. It is in conformity with the Paris Principles for the promotion and protection of human rights.

- **Statement 2 is incorrect:**

Power of the commission:

- The Commission shall have all the powers of a civil court namely:
- Summoning and enforcing the attendance of witnesses and examining them on oath;
- discovery and production of any document;
- receiving evidence on affidavits;
- requisitioning any public record or copy thereof from any court or office;
- issuing commissions for the examination of witnesses or documents;
- any other matter which may be prescribed.

- **Statement 3 is incorrect:** The Chairperson and members of the NHRC are appointed by the President of India.

Composition of the Commission:

- The National Human Rights Commission (NHRC) is composed of a Chairperson and eight other members.
- The chairperson is retired chief justice of India or a judge of the Supreme Court.
- Those eight members are:
 - Four full-time members.
 - Four deemed members.

<https://nhrc.nic.in/acts-and-rules/protection-human-rights-amendment-act-2019>

Q.59) Ans: B

Exp:

The International Labour Organisation (ILO) has released the World Employment and Social Outlook: Trends (WESO) report 2021.

About Report

- The report analyses key labour market issues including unemployment, labour underutilisation, working poverty, income inequality, labour income share and factors that exclude people from decent work.
- It highlights how the COVID-19 crisis had worsened pre-existing inequalities by hitting vulnerable workers harder.

Key Findings

- Covid-19 induced jobs loss will reach 75 million in 2021, before falling to 23 million in 2022. Unemployment is expected to stand at 205 million people in 2022.
- The quality of newly created jobs would likely deteriorate in developing and emerging economies.
- Compared to 2019, an additional 108 million workers worldwide are now categorized as poor or extremely poor.

https://www.ilo.org/global/research/global-reports/weso/2021/WCMS_795453/lang-en/index.htm

Extra Edge by Only IAS

International Labour Organisation

- It was created in 1919, as part of the Treaty of Versailles that ended World War I, to reflect the belief that universal and lasting peace can be accomplished only if it is based on social justice.
- It became a specialized agency of the United Nations in 1946.
- It is a tripartite organization, the only one of its kind bringing together representatives of governments, employers and workers in its executive bodies.
- India is a founding member of the ILO with a total 187 member States.
- Headquarter is situated at Geneva in Switzerland.
- In 1969, ILO received the Nobel Peace Prize for improving fraternity and peace among nations, pursuing decent work and justice for workers, and providing technical assistance to other developing nations.
- In 2020 India assumed the Chairmanship of the Governing Body of ILO.

Q.60) Ans: D

Exp:

- **Statement 1 is correct:** The 'One Nation One Standard' Mission was envisioned on the line of

'one nation, one ration card' scheme in order to ensure quality products in the country. The purpose is to converge multiple standards with the BIS which is a recognised national body for standardisation in India. The recognition is valid for 3 years and will require renewal after completion of the validity period.

- **Statement 2 is correct:** This mission is implemented by the Bureau of Indian Standard under the aegis of the Ministry of Consumer Affairs, Food & Public Distribution. The Bureau of Indian Standard has been established for the harmonious development of the activities of standardization, marking and quality certification of goods and for matters concerned to it.
- **Statement 3 is correct:** RDSO (Research Design & Standards Organization) became the first Institution to be declared Standard Developing Organization (SDO) under one "One Nation One Standard" mission on BIS (Bureau of Indian Standards). It Sets a template for all the rest of leading research and standard development organizations in the country to follow and adapt World Class Standards.

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1723360>

Q.61) Ans: B

Exp:

- **Statement 1 is incorrect:** The Flag Satyagraha/Jhanda Satyagraha movement by the freedom fighters shook the British government and it infused a new life into the freedom movement. It was held in Jabalpur and Nagpur in 1923. The news of flag hoisting in Jabalpur spread like fire in the country and after flags were hoisted at several places across the country.
- **Statement 2 is correct:** It is a campaign of peaceful civil disobedience that focused on exercising the right and freedom to hoist the nationalist flag and challenge the legitimacy of the British Rule in India through the defiance of laws prohibiting the hoisting of nationalist flags and restricting civil freedoms. Nationalist leaders such as Sardar Vallabhbhai Patel, Jamnalal Bajaj, Chakravarthi Rajagopalachari, Dr. Rajendra Prasad and Vinoba Bhave organised the revolt and thousands of people from different regions traveled to Nagpur and other parts of the Central Provinces to participate in civil disobedience.

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1728339>

Q.62) Ans: C

Exp:

- **Statement 1 is correct:** National Maritime Heritage Complex, a world-class facility is to be developed in the vicinity of the ASI site of Lothal, Gujarat. It will showcase and preserve India's rich and diverse maritime heritage and also display objects relating to ships and travel on large bodies of water. It also intends to highlight the ancient shipbuilding and navigational technologies developed by India.
- **Statement 2 is correct:** The Ministry of Culture (MoC) and Ministry of Ports, Shipping and Waterways (MoPSW) signed a MoU for Cooperation in Development of National Maritime Heritage Complex (NMHC). The project is being implemented by the Ministry of Shipping through its Sagarmala programme, with the involvement of the Archaeological Survey of India (ASI), the State government and other stakeholders. India and Portugal will cooperate in the setting up of a national maritime heritage museum at Lothal in Gujarat.

Extra Edge by Only IAS

About Lothal Site

- The city of Lothal stood beside a Bhogava river, a tributary of Sabarmati, in Gujarat, close to the Gulf of Khambhat.
- It was situated near areas where raw materials such as semi-precious stones were easily available.
- This was an important centre for making objects out of stone, shell and metal.
- There was also a storehouse in the city.
- Many seals and sealings (the impression of seals on clay) were found in this storehouse.
- A building that was found here was probably a workshop for making beads: pieces of stone, half made beads, tools for bead making, and finished beads have all been found here
- The excavated site of Lothal is the only port-town of the Indus Valley Civilisation.

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1727640>

Q.63) Ans: D

Exp:

- **Statement 1 is correct:** Inland Vessels Bill, 2021 will replace the Inland Vessels Act, 1917. The Act provides for the regulation of inland vessel navigation by states including the registration of vessels, and safe carriage of goods and passengers. The Inland Vessels bill proposes to streamline and regulate how vessels travel in the inland water fields of India.

- **Statement 2 is correct:**

Key features of the Bill:

- It provides for a unified law for the entire country, instead of separate rules framed by the States.
- The certificate of registration granted under

to be enrolled at district, taluk or panchayat or village level.

- **Statement 3 is correct:** More than 4000 km of the inland waterways are operational in India and with the new bill the government would get to promote the development of India's inland shipping fleet. It also expects more ports to be developed around the country to promote the use of cargo. It will also help to promote Inland Water Transport

Inland Water Transport (IWT) in India:

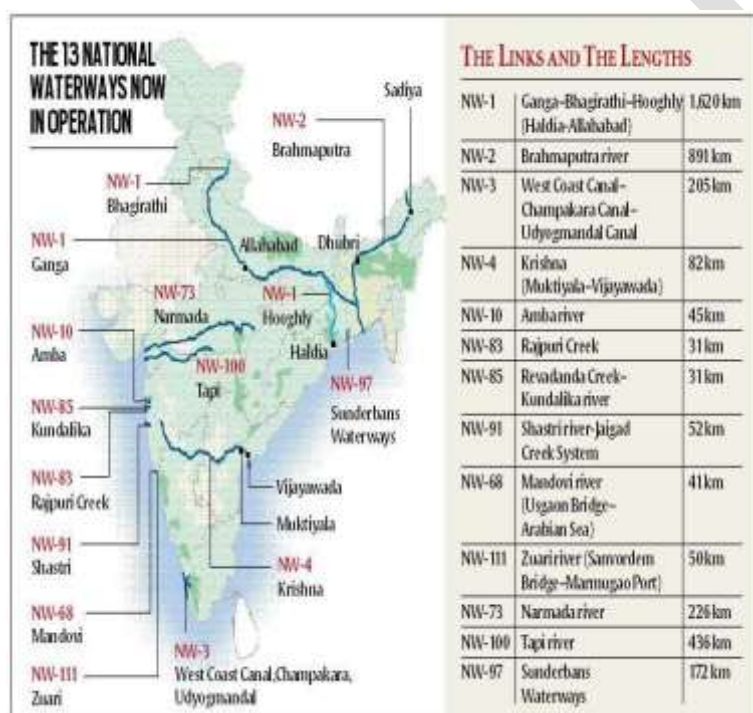
- India has about 14,500 km of navigable waterways which consist of rivers, canals, backwaters, creeks, etc.
- IWT is a fuel-efficient and environment-friendly mode.
- As per the National Waterways Act 2016, 111 waterways have been declared as National Waterways (NWs).
- The Inland Waterways Authority of India (IWAI) is implementing the Jal Marg Vikas Project (JMVP) at an estimated cost of ₹5369.18 crores for capacity augmentation of navigation on the Haldia-Varanasi stretch of Ganga (part of NW-1) with the technical and financial assistance of the World Bank.

<https://prsindia.org/billtrack/the-inland-vessels-bill-2021>

Q.64) Ans: B

Exp:

- **Statement 1 is incorrect:** National Monetisation pipeline (NMP) is launched in an effort to list out the government's infrastructure assets to be sold over the next four-years. It is prepared by NITI Aayog in consultation with infrastructure line ministries, based on the mandate for 'Asset Monetisation' under Union Budget 2021-22. It's objective is to unlock the value of investments in brownfield public sector assets by tapping institutional and long-term capital, which can thereafter be leveraged for public investments.
- **Statement 2 is correct:** The National Monetisation Pipeline comprises a four-year pipeline from 2022-2025 of the central government's brownfield Infrastructure Assets. The top three sectors identified for asset monetisation include railways, airports and coal mining. Besides these, assets such as roads, power transmission lines and gas pipelines will also be part of the monetisation plan.



the proposed law will be deemed to be valid in all States and Union Territories, and there will be no need to seek separate permissions from the States.

- The Bill provides for a central database for recording the details of vessel, vessel registration, crew on an electronic portal.
- It requires all mechanically propelled vessels to be mandatorily registered. All non-mechanically propelled vessels will also have



Source:

<https://pib.gov.in/PressReleaseDetail.aspx?PRID=1748297>

Q.65) Ans: C

Exp:

- **Statement 1 is correct:** Indian Foreign Exchange reserves hit a new lifetime high of around US\$ 612 billion, making India the fourth largest Forex reserve holder after China, Japan and Switzerland. Foreign Exchange Reserves, also known as Forex Reserves, are assets held on reserve by a central bank in foreign currencies. Mostly dominated by foreign currency assets, it can also include other instruments like bonds, treasury bills, Gold Reserves, Special Drawing Rights at IMF etc.
- **Statement 2 is correct:** Recently, US dollar, Euro, British pound, Japanese Yen and Chinese Yuan are some of the common currency assets with US dollar as main currency due to its use in settlement of all international transactions. As the central bank the Reserve Bank of India is responsible for management of Forex reserves under Reserve Bank of India Act, 1934 and Foreign Exchange Management Act, 1999.

Some of the major reasons for this sudden rise includes:

- High Foreign Capital Flow in terms of Rising Foreign Direct Investment (FDI) and Foreign Portfolio Investment (FPI) due to its large market size, growing startups, corporate tax cut, higher returns etc.
- Reduced Capital out flow due to reduced consumptions under Covid-19 and curbs on foreign travel. E.g. In 2020-21, India's BoP was at record surplus of \$87 billion.
- Record remittances from its Diaspora in the last two years (above US\$ 80 billion), Massive Liquidity injection in USA through stimulus measures against economic impact of Covid-19 pandemic. E.g. US\$2.3 trillion lending support from the Federal Reserve to households, financial markets, government etc.

Source:

<https://economictimes.indiatimes.com/markets/forex/forex-reserves-drop-by-2-47-bn-to-616-895-bn/articleshow/85688345.cms>

Q.66) Ans: C

Exp:

- Plastic Waste Management Amendment Rules, 2021 amend the 2016 rules. The manufacture, import, stocking, distribution, sale and use of the identified single-use plastic will be prohibited with effect from the 1st July, 2022.
- The ban will not apply to commodities made of compostable plastics. For banning other plastic commodities in the future, other than those that have been listed in this notification, the government has given industry ten years from the date of notification for compliance.
- The permitted thickness of the plastic bags, currently 50 microns, will be increased to 75 microns from 30th September, 2021, and to 120 microns from the 31st December, 2022. Plastic bags with higher thickness are more easily handled as waste and have higher recyclability.
- The Central Pollution Control Board, along with state pollution bodies, will monitor the ban, identify violations, and impose penalties already prescribed under the Environmental Protection Act, 1986.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1745433>

Q.67) Ans: A

Exp:

- **Statement 1 is correct:** The 'Forum for Decarbonizing Transport' will be a one-stop platform for initiating dialogues between the stakeholders and partners that will help in achieving specific results in reducing emissions from the transport sector. It will work in close coordination with all the stakeholders to formulate strategies and develop appropriate business models to accelerate electric mobility in India.
- **Statement 2 is incorrect:** In India, transport is the third most CO2 emitting sector with road transport contributing to more than 90% of total CO2 emissions. The Indian government through its various policies is promoting decarbonization of transport with a major focus on electric mobility. NITI Aayog and World Resources Institute (WRI)-India, jointly launched the 'Forum for Decarbonizing Transport' in India as part of the NDC-Transport Initiative for Asia (NDC-TIA) project.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1748514>



Q.68) Ans: A

Exp:

'surangam'

- A special water harvesting structure is called surangam. It is a horizontal well mostly excavated in hard laterite rock formations. The excavation continues until a good amount of water is struck. Water seeps out of the hard rock and flows out of the tunnel.
- This water is usually collected in an open pit constructed outside the surangam. Surangams are similar to qantas which once existed in Mesopotamia and Babylon around 700 Before Common Era. This system has been very effectively used for domestic and agriculture purposes in dry areas of northern Malabar. It is usually found in northern Kerala and southern Karnataka.

Source: <https://www.downtoearth.org.in/news/water/fate-of-afghanistan-s-karez-system-uncertain-but-south-india-s-surangam-thriving-78584>

Q.69) Ans: A

Exp:

- **Statement 1 is correct** : The first National Heart Failure Biobank (NHFB) in the country that would collect blood, biopsies, and clinical data as a guide to future therapies was inaugurated at the Sree Chitra Tirunal Institute for Medical Sciences and Technology (SCTIMST). Biobank will provide insights into heart diseases and heart failure among Indian children and adults.
- **Statement 2 is correct** : The biosamples include the blood, serum, tissue samples obtained during open-heart surgery and peripheral blood mononuclear cells (PBMCs) and genomic DNA collected from heart failure patients.
- **Statement 3 is incorrect** : The **biobank activity is supervised by a Technical Advisory Committee (TAC) with a member from ICMR**. The samples are collected after informed consent from patients who are willing to donate specimens. The samples stored and catalogued will be linked to clinical data such as physiological measures, imaging data like ECG, Echocardiography MRI, and follow-up data.

Source - <https://dst.gov.in/india-first-biobank-heart-failure-research-inaugurated-sree-chitra-tirunal-institute-medical>

Q.70) Ans: B

Exp:

- The **Pensilungpa Glacier located in Ladakh's Zaskar Valley** is retreating due to increase in

temperature and decrease in precipitation during winters.

- Since 2015, the Wadia Institute of Himalayan Geology (WIHG) at Dehradun, an autonomous body under the Department of Science and Technology, has been working on various aspects on glaciology – glacier health (mass balance) monitoring, dynamics, discharge, past climatic conditions, speculation for future climate change and its impact on glaciers.

Source -

<https://www.thehindu.com/sci-tech/science/ladakh-glacier-retreat-due-to-warming-low-winter-precipitation-finds-study/article35788810.ece/amp/>

Q.71) Ans: A

Exp:

Stake networking

- The Pensilungpa Glacier located in Ladakh's Zaskar Valley is retreating due to increase in temperature and decrease in precipitation during winters, a recent study has found.
- "Based on field observations for glacier mass balance collected via stake networking over the glacier surface from 2016-2019, they assessed the impact of climate change through the lens of past and present response of the Pensilungpa Glacier, Zaskar Himalaya, Ladakh.
- **In stake networking, a stake made of bamboo, is installed on the glacier surface using the steam drill for mass balance measurement.**
- The study also points at the significant influence of debris cover on the mass balance and retreat of the glacier's endpoint, especially in summer. Furthermore, the mass balance data for the last 3 years (2016–2019) showed a negative trend with a small accumulation area ratio.

Source - <https://www.thehindu.com/sci-tech/science/ladakh-glacier-retreat-due-to-warming-low-winter-precipitation-finds-study/article35788810.ece/amp/>

Q.72) Ans: A

Exp:

- **Statement 1 is correct** : ReCAAP is the first regional government-to-government agreement to promote and enhance cooperation against piracy and armed robbery against ships in Asia. The regional body has its headquarters in Singapore.



- **statement 2 is incorrect** : The agreement was launched in November 2006 with 14 Asian contracting parties. The number has increased to 21 today (Aug, 2021), including European countries like Norway, the Netherlands, Denmark, and the United Kingdom, Australia, and the United States.

Source -

<https://www.hindustantimes.com/india-news/india-defeats-china-for-recaap-elections-with-quad-on-its-side-101628177963037-amp.html>

Q.73) Ans: C

Exp:

- Women scientists from the Indian Agricultural Research Institute (IARI) of ICAR developed Refrigeration System Pusa-FSF & a technology related to extraction of gluten from wheat dough and its regeneration. The 10x10x10 ft cold store structure designed by the scientists can stock enough vegetables and fruits to fully load a Tata 407 mini truck

It is a 100% solar-powered battery-less cold store that can preserve around 2 tonnes of freshly-harvested produce at 3-4°C during the day and 8-12°C at night time.

It minimises the post-harvest losses, by having a cold store at the farm itself and running on solar power without any electricity from the grid.

Source: <https://journalsofindia.com/refrigeration-system-pusa-fsf/>

Q.74) Ans: A

Exp:

- **Statement 1 is correct:** Rashtriya Krishi Vikas Yojana - Remunerative Approaches for Agriculture and Allied sector Rejuvenation (RKVY-RAFTAAR) with broad objectives of making farming a remunerative economic activity through strengthening the farmer's effort, risk mitigation with major focus on development and creation of pre & post-harvest infrastructure.

MANAGE-CIA, Centre for Innovation and Agripreneurship is a Centre of Excellence and Knowledge Partner to Ministry of Agriculture and Farmers' Welfare for Implementation of RKVY-RAFTAAR Scheme. As a knowledge partner, MANAGE will handhold and demonstrate best practices to RKVY-RAFTAAR Agribusiness Incubators (R-ABI) which includes Indian institute of Millet Research (IIMR), Tamil Nadu Agriculture University (TNAU), Acharya N. G. Ranga Agricultural University (ANGRAU) and Kerala Agriculture University (KAU).

- **Statement 2 is incorrect:** Under this scheme, funds are released to the State Governments/UTs on the basis of projects related to agriculture and allied sectors, approved by the State Level Sanctioning Committee (SLSC) headed by the Chief Secretary of the concerned State/UT which is the empowered body to approve projects for implementation under this scheme. There is an Agri-Entrepreneurship & innovation component in RKVY-RAFTAAR which is operated by the Department of Agriculture, Cooperation & Farmers Welfare.

Source:

<https://pib.gov.in/PressReleaseDetailm.aspx?PRID=1697506>

Q.75) Ans: D

Exp:

- Atal Innovation Mission (AIM), NITI Aayog, launched the Atmanirbhar Bharat ARISE-Atal New India Challenges, to spur applied research and innovation in Indian MSMEs and startups.
- The programme will be driven by Indian Space Research Organization (ISRO), four ministries—Ministry of Defence; Ministry of Food Processing Industries; Ministry of Health and Family Welfare; and Ministry of Housing and Urban Affairs—and associated industries to facilitate innovative solutions to sectoral problems
- The objective of Atmanirbhar Bharat ARISE-ANIC program is to proactively collaborate with esteemed Ministries and the associated industries to catalyse research, innovation and facilitate innovative solutions to sectoral problems. The objective is also to provide a steady stream of innovative products & solutions where the Central Government Ministries / Departments will become the potential first buyers. The Atmanirbhar Bharat ARISE-ANIC program is in line with the Honourable Prime Minister's mandate of "Make in India", "Startup India", "Atmanirbhar Bharat" to fast track the growth of the Indian MSME sector. Both the MSMEs and Startups can apply for the Atmanirbhar Bharat ARISE- ANIC initiative

Source:

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1652751> and <https://aim.gov.in/arise-who-can-apply.php>.

Q.76) Ans: C

Exp:

- **Statement 1 is correct :** Biotechnology Industry Research Assistance Council (BIRAC) is a not-for-profit Section 8, Schedule B, Public Sector Enterprise, set up by Department of Biotechnology (DBT), Government of India as an Interface Agency to strengthen and empower the emerging Biotech enterprise to undertake strategic research and innovation, addressing nationally relevant product development needs.

Vision: "To Stimulate, foster and enhance the strategic research and innovation capabilities of the Indian biotech industry, particularly start-ups and SME's, for creation of affordable products addressing the needs of the largest section of society".

- **Statement 2 is correct :** The National Biopharma Mission (NBM) is an industry-academia collaborative mission for accelerating biopharmaceutical development in the country being implemented by the Biotechnology Industry Research Assistance Council (BIRAC).

BIRAC is an industry-academia interface and implements its mandate through a wide range of impact initiatives, be it providing access to risk capital through targeted funding, technology transfer, IP management and handholding schemes that help bring innovation excellence to the biotech firms and make them globally competitive.

Source: https://birac.nic.in/desc_new.php?id=89

Q.77) Ans: B

Exp:

- **Statement 1 is incorrect:** The Union Ministry of Textiles launched the Certified Jute Seeds Distribution Plan and Jute Farmers Awareness Workshop, organised at ICAR-CRIJAF Institute at Barrackpore, West Bengal, to urge the farmers to use certified seeds to increase their income and productivity and contribute to the country in Jute diversified and Technical textile products.
- **Statement 2 is correct:** The Improved Cultivation and Advanced Retting Exercise for Jute (Jute – ICARE) was launched in 2015 to popularize/introduce some of the better agronomic practices and recently developed microbial assisted retting among farmers intensively in a few blocks in West Bengal and Assam on pilot basis. The improved agronomic practices included -- line sowing of jute using seed drill to increase yield by 10-15%; weed management in jute by wheel hoeing/nail weeder instead of hand weeding for reducing the cost of

weeding; and distribution of quality certified seeds at 50% subsidy.

- **ICAR-Central Research Institute for Jute and Allied Fibers (ICAR-CRIJAF)**, established in 1953 as Jute Agricultural Research Institute (JARI), is only research institute on jute & allied fibres crops in India and is a constituent unit of Indian Council of Agricultural Research (ICAR) an Autonomous Body under Department of Agricultural Research & Education (DARE) under Ministry of Agriculture and Farmers Welfare Government of India. This institute is located in Barrackpore, Kolkata.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1698178> and

<https://pib.gov.in/PressReleaselframePage.aspx?PRID=1497089>

Q.78) Ans: D

Exp:

- The **Protection of Plant Variety and Farmers Right Act, 2001 (PPVFR Act)** is an Act of the Parliament of India that was enacted to provide for the establishment of an effective system for protection of plant varieties, the rights of farmers and plant breeders, and to encourage the development and cultivation of new varieties of plants
- ITPGRFA also known as Seed Treaty, is a comprehensive international agreement for ensuring food security through the conservation, exchange and sustainable use of the world's plant genetic resources for food and agriculture (PGRFA), as well as the fair and equitable benefit sharing arising from its use. It also recognizes farmers' rights, subject to national laws.
- The Intellectual Property Right granted under PPV & FR Act, 2001 is a dual right – one is for the variety and the other is for the denomination assigned to it by the breeder. The rights granted under this Act are heritable and assignable and only registration of a plant variety confers the right. Farmers are entitled to save, use, sow, re-sow, exchange or sell their farm produce including seed of a registered variety in an unbranded manner. Farmers' varieties are eligible for registration and farmers are totally exempted from payment of any fee in any proceedings under this Act. Farmers can claim for compensation if the registered variety fails to provide expected performance under given conditions. The rights granted under this Act are exclusive right to produce, sell, market, distribute, import and export the variety.



Source:

<https://www.thehindu.com/news/national/plant-varieties-regulator-revising-faqs-after-farmers-call-foul/article30271561.ece> and
<https://pib.gov.in/PressReleaseDetail.aspx?PRID=1591329>

Q.79) Ans: D

Exp:

- **Statement 1 is incorrect:** National Bee Board (NBB), a registered society under the Societies Registration Act XXI of 1860, was reconstituted by the Department of Agriculture & Cooperation, Ministry of Agriculture and Farmers Welfare, Govt. Of India with public private partnership in June, 2006.

National Bee Board (NBB) envisages overall development of Beekeeping by promoting Scientific Beekeeping in India to increase the productivity of crops through pollination and increase the Honey production for increasing the income of the Beekeepers/ Farmers.

- **Statement 2 is incorrect:** Secretary (A C & FW), Deptt. of Agriculture, Cooperation and Farmers Welfare, Ministry of Agriculture and Farmers Welfare is the ex- officio chairman of the NBB. A Beekeeping Development Board also functioned under the Chairpersonship of Secretary (A & C) to coordinate the Beekeeping activities.

Source: https://allbankcare.in/nbb/documents/NBB-Registration-Guidelines_EN.pdf and
<https://nbb.gov.in/pdf/GoverningCouncil.pdf>

Q.80) Ans: B

Exp:

- It is a six Months Mentoring Program promoting agri start-ups with innovative ideas between India & Brazil. India and Brazil are the growing economies and agriculture forms the backbone of both economies. Innovation and entrepreneurship in the agri sector are one of the key reasons for their growth, thereby promoting agri start-ups with innovative ideas. To foster excellent innovative research and development facilities across borders "Maitri" was introduced to harness the best of talent, technologies and promote technology transfer & market access across both nations for better economic growth and strong bilateral relations.
- The Pusa Krishi Incubator, of ICAR- Indian Agricultural Research Institute had launched the

"Maitri - Indo-Brazil Agri-Tech Cross Border Incubation Programme" at the National Agricultural Science Centre Complex, provide agritech start-ups access to the global market through mutual benefits and shared learning.

- The agritech startups were selected on the basis of Startup India guidelines, where the products or services offered by the startups are commercially viable. The selected startups will be a part of a six-month programme in India and Brazil, where they will get to experience workshops, networking events and mentorship opportunities, as well as pitch ideas to investors and other stakeholders in both countries.

Source: <https://www.icar.org.in/content/maitri-indo-brazil-agri-tech-cross-border-incubation-programme-launched>

Q.81) Ans: C

Exp:

- **Statement 1 is correct :** The National Bee Board (NBB), Ministry of Agriculture and Farmers Welfare under National Beekeeping & Honey Mission (NBHM) launched the Madhukranti Portal for Traceability of Source of Honey and Other Hive Products with the help of a digital platform. It has the provision of online registration to achieve a traceability source of Honey and other beehive products. An initiative to leverage high quality Honey Production meeting international standards. All Beekeepers and Entrepreneurs dealing in honey production, processing, packaging, marketing and exporting are requested to register on the portal and provide authentic business-related information for achieving the objective set by the National Bee Board under the NBH Mission

The portal enables consumers to know the source of honey and assure quality of the products.

- **Statement 2 is correct :** The technical and banking partner for development of this digital platform is Indian Bank. An MoU between National Bee Board (NBB) and Indian Bank was signed for this project.

Source:

[https://pib.gov.in/PressReleaseframePage.aspx?PRID=1710160#:~:text=Union%20Minister%20for%20Agriculture%20and,%26%20Honey%20Mission%20\(NBHM\)](https://pib.gov.in/PressReleaseframePage.aspx?PRID=1710160#:~:text=Union%20Minister%20for%20Agriculture%20and,%26%20Honey%20Mission%20(NBHM))

Q.82) Ans: C

Exp:



- **Statement 1 is incorrect:** The Central Pollution Control Board (CPCB) has developed a Comprehensive Environmental Pollution Index (CEPI). CPCB has done a nation wide environmental assessment of Industrial Clusters based on CEPI and 43 such industrial clusters having CEPI greater than 70, on a scale of 0 to 100, have been identified as critically polluted.

The data refers to the Comprehensive Environmental Pollution Index (CEPI) scores of the critically polluted industrial clusters/areas. The index captures the various dimensions of the environment including air, water and land.

Comprehensive Environmental Pollution Index (CEPI), which is a rational number to characterize the environmental quality at a given location following the algorithm of source, pathway and receptor have been developed

- **Statement 2 is incorrect:** CEPI provides comprehensive information on the extent and types of pollution caused by a particular industrial cluster, which then can be used to identify appropriate pollution mitigation measures. The source-pathway-receptor approach takes into account the complete geography that is affected by a polluting industry. As this index is a quantitative approach to understanding certain qualitative impacts caused by the polluting industry, monitoring authorities should monitor pollution and maintain proper records. With increasing industrial activity in India, Cumulative Impact Assessment is gaining attention. CEPI is an effective tool to concisely represent the wide and diversified effects of pollution using numbers.

Source: <https://cpcb.nic.in/comprehensive-environmental-pollution-index-cepi/>

Q.83) Ans: D

Exp:

- The Government of India has instituted a scheme for labelling of environment friendly products to be known as ECO Mark. The scheme is being administered by the Bureau of Indian Standards.
- The Scheme is being operated on a national basis and provides certification and labelling for household and other consumer products which meet certain environmental criteria along with quality requirements prescribed in relevant Indian Standards for the product
- The Ecomark Scheme covers various product categories like Soaps and Detergents, Paints, Food Items, lubricating oils, Packaging

materials/Package, Architectural Paints and Powder Coatings, Batteries, Electrical and electronic goods, Food Additives, Wood Substitutes, Cosmetics, Aerosols and Propellants, Plastic Products, Textiles, Fire-extinguisher, Leather and Coir & Coir Products

Source: <https://www.bis.gov.in/index.php/product-certification/operation-of-eco-mark-scheme/>

Q.84) Ans: C

Exp:

- **Statement 1 is correct :** Rapid urbanization and development has led to the degradation of the environment in urban areas. USERS has been created as a new 'pressure mechanism', which highlights the issue and facilitates information sharing on the subject thereby enabling adoption of better standards/practices. The project aims to develop an analytical tool to measure the performance with respect to basic services delivery in local bodies (Delhi Jal Board, Municipal Corporation of Delhi, Kanpur Jal Sansthan and Kanpur Nagar Nigam) of Delhi and Kanpur, identified as pilot cities
- **Statement 2 is correct :** The Urban Services Environmental Rating System (USERS) Project funded by UNDP, executed by the Ministry of Environment Forest and Climate Change and implemented by Tata Energy Research Institute. The Performance Measurement (PM) tool was developed through a set of performance measurement indicators that are benchmarked against set targets using the input-output efficiency outcome framework.

Source: <http://moef.gov.in/wp-content/uploads/report/0203/chap-09.htm> and <https://timesofindia.indiatimes.com/city/delhi/teri-rating-for-civic-agencies/articleshow/45908810.cms>

Q.85) Ans: A

Exp:

- **Statement 1 is correct :** The PUSA KRISHI mobile app helps the farmers to find easy solutions to problems in their farm fields and get information about weather and accordingly take measures to save crops. It also offers information related to new varieties of crops developed by Indian Council of Agriculture Research (ICAR), resource conserving cultivation practices as well as farm machinery and its implementation will help in increasing returns to farmers. Details of 185



technologies are available on Pusa Krishi Mobile App

- **Statement 2 is incorrect** : Pusa Krishi Mobile App is available in two languages viz., Hindi and English

Kisan Suvidha Mobile application which was available in 9 languages viz. English, Hindi, Tamil, Gujarati, Oriya, Punjabi, Marathi, Bangla and Telugu which gives information on the weather of the current day and next 5 days, dealers, market prices, agro advisories, plant protection, IPM Practices, etc.

Q.86) Ans: D

Exp:

The World Meteorological Organization (WMO) released a report titled the State of Climate Services report 2021. It focuses on Terrestrial Water Storage.

Terrestrial Water Storage (TWS):

- TWS is the sum of all water on the land surface and in the subsurface, i.e. surface water, soil moisture, snow and ice and groundwater.
- Water resources across the world are under tremendous pressure due to human and naturally-induced stressors.
- These include population growth, urbanisation and decreasing availability of freshwater.
- Extreme weather events too have been responsible for the pressure on water resources realised across sectors and regions

Global Scenario:

- TWS dropped at a rate of 1 cm per year in 20 years (2002-2021).
- The biggest losses have occurred in Antarctica and Greenland. But many highly populated, lower latitude locations have also experienced TWS losses.

Indian Scenario:

- The TWS has been lost at a rate of at least 3 cm per year. In some regions, the loss has been over 4 cm per year too.
- India has recorded the highest loss in terrestrial water storage if the loss of water storage in Antarctica and Greenland is excluded.
- India is the 'topmost hotspot of TWS loss'. The northern part of India has experienced the maximum loss within the country.

World Meteorological Organization(WMO)

- WMO is an intergovernmental organization with a membership of 192 Member States and Territories.
- It originated from the International Meteorological Organization (IMO), which was established after

the 1873 Vienna International Meteorological Congress.

- Established by the ratification of the WMO Convention on 23 March 1950, WMO became the specialized agency of the United Nations for meteorology (weather and climate), operational hydrology and related geophysical sciences.'
- It's Headquartered at Geneva, Switzerland.

Source: <https://reliefweb.int/report/world/2020-state-climate-services-move-early-warnings-early-action>

Q.87) Ans: A

Exp:

- Many of the big questions in the social sciences deal with cause and effect. How does immigration affect pay and employment levels? How does a longer education affect someone's future income? These questions are difficult to Ans: because we have nothing to use as a comparison. We do not know what would have happened if there had been less immigration or if that person had not continued studying.
- However, this year's Laureates have shown that it is possible to Ans: these and similar questions using natural experiments. The key is to use situations in which chance events or policy changes result in groups of people being treated differently, in a way that resembles clinical trials in medicine.
- Using natural experiments, David Card has analysed the labour market effects of minimum wages, immigration and education. His studies from the early 1990s challenged conventional wisdom, leading to new analyses and additional insights. The results showed, among other things, that increasing the minimum wage does not necessarily lead to fewer jobs. We now know that the incomes of people who were born in a country can benefit from new immigration, while people who immigrated at an earlier time risk being negatively affected. We have also realised that resources in schools are far more important for students' future labour market success than was previously thought.

Source: <https://www.nobelprize.org/prizes/economic-sciences/2021/press-release/>

Q.88) Ans: D

Exp:

He is also known as "Lok Nayak". People also affectionately called him JP. He was a freedom fighter and one of the pioneers of the socialist movement in



India. He was a man of great intellect and ethical values and standards.

Contributions of JP Narayan:

He worked relentlessly for the poor and the underprivileged and became a symbol in the fight against corruption, anti-democratic conduct.

He fought against the repressive practices of Indira Gandhi's government Bhoodan in the 1970s. Furthermore, he also created support for the movement. However, his life had a transition from socialism to Sarvodaya. This also led to the initiation of the Total Revolution. **Total Revolution**

It is a revolution against the Indira Gandhi Regime, as she was found guilty of violating electoral laws by the Allahabad High Court. JP advocated a program of social transformation which he termed 'Sampoorna (total revolution) in 1974 against corruption in public life. This resulted in the defeat of the Indira Gandhi government in 1977 and the installation of the first-ever non-Congress government at the Centre. What is so great was his stature, the position President of Prime minister or was within his reach, but he chose to work for the people at the grassroots level. This made him a true Lok Nayak. Jayaprakash Narayan was posthumously conferred with India's highest civilian award, the Bharat Ratna in 1999 .

Source:

<https://www.britannica.com/biography/Jayaprakash-Narayan>

Q.89) Ans: A

Exp:

- The International Energy Agency is a Paris-based autonomous intergovernmental organisation established in the framework of the Organisation for Economic Co-operation and Development (OECD) in 1974 in the wake of the 1973 oil crisis.
- The IEA was initially dedicated to responding to physical disruptions in the supply of oil, as well as serving as an information source on statistics about the international oil market and other energy sectors.
- It is best known for the publication of its annual World Energy Outlook.
- It has 30 members at present. The International Energy Agency family also includes eight association countries. A candidate country must be a country of the OECD (1961). But all OECD members are not IEA members.

- India became an Associate member of IEA in March 2017 but it was in engagement with IEA long before its association with the organization.

Source:

<https://www.bloomberg.com/news/articles/2021-10-14/global-gas-crisis-is-spilling-over-into-oil-markets-iea-says>

Q.90) Ans: D

Exp:

- Plastic packaging, as per the rules divided into three categories.
 - Category 1: "Rigid" plastic
 - Category 2: "Flexible plastic packaging of a single layer or multilayer (more than one layer with different types of plastic), plastic sheets and covers made of plastic sheet, carry bags or pouches.
 - Category 3: It is called multi-layered plastic packaging, which has at least one layer of plastic and at least one layer of material other than plastic.
- Producers of plastic would be obliged to declare to the government, via a centralised website, how much plastic they produce annually.
- Companies would have to collect at least 35% of the target in 2021-22, 70 %by 2022-23 and 100
- In 2024, a minimum of 50% of their rigid plastic (category 1) would have to as would 30% of their category 2 and 3 plastics.
- Every year would see progressively higher targets and after 2026-27, 80% of their category 1 and 60% of the other two categories would need to be recycled.
- If entities cannot fulfil their obligations, they would on a "case basis by case" be permitted to buy certificates making up for their shortfall from organisations that have used recycled content in excess of their obligation.
- The CPCB would develop a "mechanism" for such a portal.
- Non-compliance would not invite a traditional fine. Instead, an "environmental compensation" would be levied, though the rules do not specify how much this compensation would be.

Source: <https://www.thehindu.com/sci-tech/energy-and-environment/centre-announces-plastic-waste-recycling-targets/article36973023.ece>

Q.91) Ans: b

Exp:



Fluorinated Gases

- In general, fluorinated gases are the most potent and longest lasting type of greenhouse gases emitted by human activities. There are four main categories of fluorinated gases—hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulphur hexafluoride (SF₆), and nitrogen trifluoride (NF₃)
- Fluorinated greenhouse gases (F-gases) are a family of gases containing fluorine. They are powerful greenhouse gases that trap heat in the atmosphere and contribute to global warming. They are stronger than naturally occurring greenhouse gases and their use is regulated.
- Like other long-lived greenhouse gases, fluorinated gases are well-mixed in the atmosphere, spreading around the world after they're emitted. Fluorinated gases are removed from the atmosphere only when they are destroyed by sunlight in the far upper atmosphere.

Source: <https://www.icac.com/page/Greenhouse>

Q.92) Ans: D

Exp:

Climate forcing

- Climate forcing is the physical process of affecting the climate on the Earth through a number of forcing factors. These factors are specifically known as forcings because they drive the climate to change, and it is important to note that these forcings exist outside of the existing climate system.
- The climate system includes the hydrosphere, land surface, the cryosphere, the biosphere, and atmosphere. Examples of some of the most important types of forcings include: variations in solar radiation levels, volcanic eruptions, changing albedo, and changing levels of greenhouse gases in the atmosphere.

Each of these are considered external forcings because these events change independently of the climate, perhaps as a result of changes in solar activity or human-caused fossil fuel combustion.

Q.93) Ans: b

Exp:

- **Statement 1 is incorrect:** The Global Warming Potential (GWP) was developed to allow comparisons of the global warming impacts of different gases and not only the GHGs. Specifically, it is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO₂).
- **Statement 2 is correct:** Gases with a higher GWP absorb more energy, per pound, than gases with a lower GWP, and thus contribute more to warming earth.

Carbon dioxide with a GWP of '1' serves as a baseline for other GWP values. The larger the GWP, the more the gas causes. For example, Methane's 100-year GWP is 21, which means methane will cause 21 times as much warming as an equivalent mass of carbon dioxide over a 100- year time period.

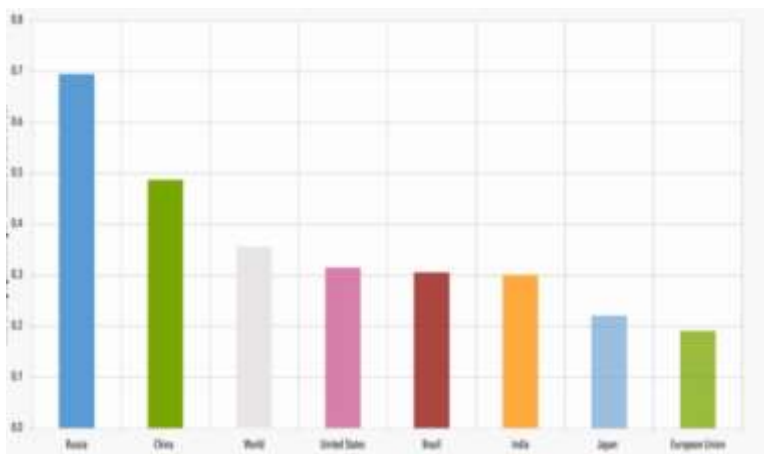
Q.94) Ans: a

Exp:

- **Statement 1 is correct:** Greenhouse gas intensity is a ratio comparing the greenhouse gas (GHG) emissions of an activity or economic sector to the economic value it generates. In recent years, many countries have calculated the GHG intensity of their economic sectors as a basis for policy making.

GHG intensity = (GHGs released in energy consumption for production. + GHGs released in energy consumption for overhead. + GHGs released by transport used for business travel. + Additional GHGs released from production process).

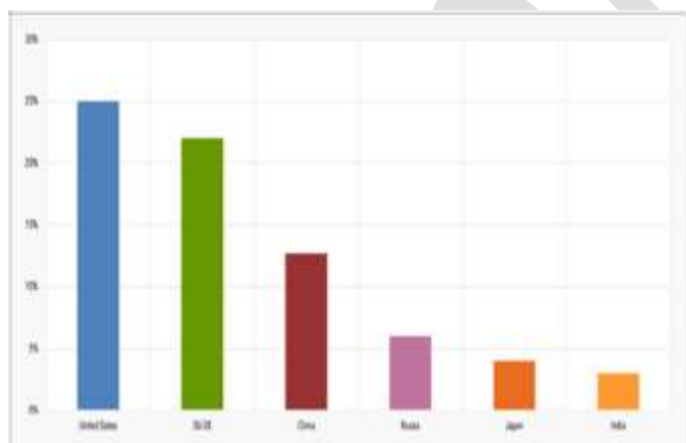
- **Statement 2 is correct:** It is highest in Russia and China with the United States below the world average.



Q.95) Ans: a

Exp:

- **Statement 1 is correct:** Cumulative Emission is the level of GHG gases concentration in the atmosphere, not the current level of emissions, which determines the extent of the warming effect from their presence in the atmosphere. In other words, it is the cumulative effect of past emissions from all sources that matters – the stock of CO₂ rather than the flow. This is based on territorial emissions, which do not account for emissions embedded in traded goods.
- **Statement 2 is incorrect:** The United States of America tops the chart, followed by the European Union and China.



Cumulative Emissions 1751- 2017

Q.96) Ans: d

Exp:

Basel Convention

- The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal was adopted in 1989 and it came into force in 1992. It is the most comprehensive global environmental agreement on hazardous wastes and other wastes.
- The Convention aims to protect human health and the environment against the adverse effects resulting from the generation, transboundary movements and management of hazardous wastes and other wastes.
- The Convention covers toxic, poisonous, explosive, corrosive, flammable, ecotoxic and infectious wastes. Parties also have an obligation to minimize the quantities that are transported, to treat and dispose of wastes as close as possible to their place of generation and to prevent or minimize the generation of wastes at source. Examples of wastes regulated by the Basel Convention:

- Biomedical and Healthcare wastes
- Used oils
- Used lead batteries
- Persistent Organic Pollutant Wastes
- Polychlorinated Biphenyls
- Chemical wastes generated by industries and other consumers.

Source:

<https://www.unep.org/resources/report/basel-convention-control-transboundary-movements-hazardous-wastes>

Q.97) Ans: c

Exp:

- **Eutrophication**, the gradual increase in the concentration of phosphorus, nitrogen, and other plant nutrients in an aging aquatic ecosystem such as a lake. The productivity or fertility of such an ecosystem naturally increases as the amount of organic material that can be broken down into nutrients increases.

• The most common nutrients causing eutrophication are nitrogen N and phosphorus P. The main source of nitrogen pollutants is run-off from agricultural land, whereas most phosphorus pollution comes from households and industry, including phosphorus-based detergents.

• **Option A relates to the process of**

Bioaccumulation, which is the gradual accumulation of substances, such as pesticides or other chemicals, in an organism. Bioaccumulation occurs when an organism absorbs a substance at a rate faster than that at which the substance is lost or eliminated by catabolism and excretion.

• **Option B related to the process of**

Biomagnification, also known as bio amplification or biological magnification, is any concentration of a toxin, such as pesticides, in the tissues of tolerant organisms at successively higher levels in a food chain.

Q.98) Ans: d

Exp:

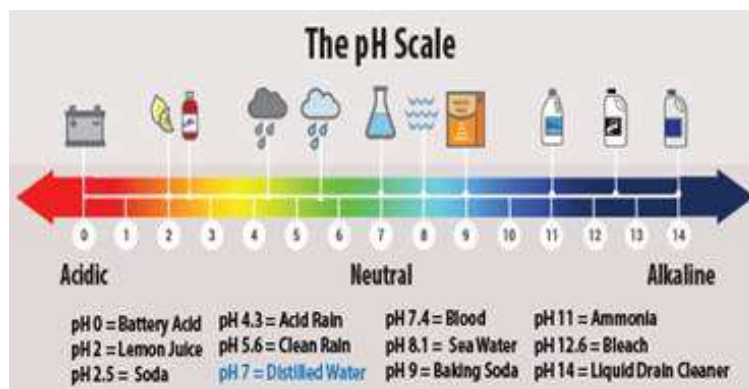
About Acid Rain

• Acid rain results when **sulfur dioxide (SO₂)** and **nitrogen oxides (NO_x)** are emitted into the atmosphere and transported by wind and air currents. The **SO₂ and NO_x react with water**, oxygen and other chemicals to **form sulfuric and nitric acids**. These then mix with water and other materials before falling to the ground.

• Winds can blow SO₂ and NO_x over long distances and across borders making acid rain a problem for everyone and not just those who live close to these sources.

• While a small portion of the SO₂ and NO_x that cause acid rain is from natural sources such as volcanoes, most of it comes from the burning of fossil fuels. The major sources of SO₂ and NO_x in the atmosphere are:

• Burning of fossil fuels to generate electricity.



• Two thirds of SO₂ and one fourth of NO_x in the atmosphere come from electric power generators.

• Vehicles and heavy equipment.

• Manufacturing, oil refineries and other industries.

• Acidity and alkalinity are measured using a pH scale for which 7.0 is neutral. The lower a substance's pH (less than 7), the more acidic it is; the higher a substance's pH (greater than 7), the more alkaline it is. Normal rain has a pH of about 5.6; it is slightly acidic because carbon dioxide (CO₂) dissolves into it forming weak carbonic acid. Acid rain usually has a pH between 4.2 and 4.4.

Q.99) Ans: a

Exp:

Statement 1 is correct: Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal. It is an **international treaty that aims to reduce the movement of hazardous waste between countries**.

- **Statement 2 is incorrect: Basel agreement is not legally binding.** It particularly focuses on preventing transfer of hazardous waste from developed to less developed countries.
- Recently, the 14th meeting of the Basel Convention, which lays down guidelines on the movement of hazardous waste, concluded in Geneva on the 10th May, 2019 after two weeks of negotiations involving 187 countries.
- **The Geneva meeting amended the 1989 Basel Convention on the control of hazardous wastes to include plastic waste in a legally-binding**



framework. The new amendment would empower developing countries to refuse “**dumping plastic waste**” by others.

- The meeting also undertook to eliminate two toxic chemical groups — **Dicofol and Perfluorooctanoic Acid**, plus related compounds. The latter has been used in a wide variety of industrial and domestic applications including non-stick cookware and food processing equipment, as well as carpets, paper and paints.

Q.100) Ans: a

Exp:

- **India is testing a Swedish technology — torrefaction that can convert rice stubble into ‘bio-coal’.** The Office of the Principal Scientific Advisor (PSA) to Government of India has funded a pilot project in Punjab to evaluate the feasibility of the technology. Bioendev, a Swedish company, has set up a pilot plant at the National Agri-Food Biotechnology Institute in Mohali.
- It is a **thermal process used to produce high-grade solid biofuels from various streams of woody biomass or agro residues.**
- The end product is a stable, homogeneous, high quality solid biofuel with far greater energy density and calorific value than the original feedstock, providing significant benefits in logistics, handling and storage, as well as opening up a wide range of potential uses.